Scalable Musculoskeletal Model for Dynamic Simulations of Upper Body Movement

Ali Nasr[a], Arash Hashemi[a], and John McPhee[a]

[a]Department of Systems Design Engineering, University of Waterloo, Canada

ABSTRACT

A musculoskeletal (MSK) model is a valuable tool for assessing complex biomechanical problems, estimating joint torques during motion, optimizing motion in sports, and designing exoskeletons and prostheses. This study proposes an open-source upper body MSK model that supports biomechanical analysis of human motion. The MSK model of the upper body consists of 8 body segments (torso, head, left/right upper arm, left/right forearm, and left/right hand). The model has 20 degrees of freedom (DoFs) and 40 muscle torque generators (MTGs), which are constructed using experimental data. The model is adjustable for different anthropometric measurements and subject body characteristics: sex, age, body mass, height, dominant side, and physical activity. Joint limits are modeled using experimental dynamometer data within the proposed multi-DoF MTG model. The model equations are verified by simulating the joint range of motion (ROM) and torque; all simulation results have a good agreement with previously published research.

KEYWORDS

Musculoskeletal system model; Dynamic simulation; Biomechanics; Muscle torque; Upper body

1. Introduction

An musculoskeletal (MSK) model is a valuable tool for evaluating complicated biomechanical problems [1], simulating and evaluating injuries [2, 3], designing and controlling exoskeletons [4–6], prostheses [7], and rehabilitation robots [8], estimating joint torques in a certain posture or during motion [2], predicting the outcome of orthopedic procedures [2, 9], optimizing motion or tools in sports engineering [10], and studying the central nervous system (CNS) for optimal control [2, 11–13]. The mechanical complexity of the musculoskeletal system necessitates the use of computer models that are both sufficiently accurate and reasonably efficient [3, 8, 14].

Skeletal and muscular elements are the two subsystems that make up the MSK system [3, 15]. Specifically, the bones and the connective tissues (such as cartilages and ligaments) make up the skeletal system. Skeletal muscles are similarly organized in layers over the bones, with tendons connecting them to the bones so that the forces created by the contractile components of the muscle fibers drive the body motions [1, 2, 16]. The activation of each muscle group causes one or more joints to move [12].

Most MSK models are mainly used for biomechanical simulation [3, 9] and not biomechatronics, i.e. they cannot be readily utilized for mechanical tool design prob-

CONTACT A. Nasr Author. Email: a.nasr@uwaterloo.ca

lems that involve human-tool interaction (e.g., golf clubs [10], robotic mechanisms [6], exoskeletons [5, 17], or bicycles [18]). The objective of this utilization is the biomechanical analysis and synthesis stage of designed mechanisms [2, 9], deriving a muscle-driven model for developing control frameworks for wearable robotic systems [4, 8], and integrating sensor information (e.g., surface electromyography (sEMG), electroencephalography (EEG), inertial measurement unit (IMU), camera) into control algorithms [4]. In addition, it is critical to have an MSK model for real-time data collection, simulation, and visualization in virtual reality applications with the patient in the loop [19]. One of the objectives of this paper is to develop a user-friendly adjustable model to aid biomechanical analysis, incorporate external sensory data, investigate various optimal control algorithms for the CNS, and optimize the design of sports tools, rehabilitation robots, and assistive devices.

Biomechanical MSK models are limited by measurement processes and the population on which they are built [3, 20–22]. As a result, the model should not be used or studied outside of its intended use [23]. So far, multivariate analyses have been performed to assess the effect of subject characteristics on peak isometric and isokinetic strength. Consequently, multiple regression equations have been proposed for the strength prediction from anthropometric values [24]. These values, along with subject body characteristics, on which the biomechanical musculoskeletal model relies, consist of sex, mass, stature, age, dominant side, and physical activity; in particular, sex, body mass, and body height impact the anatomical locations, the center of mass (COM), and inertia parameters [25, 26]. Furthermore, sex [27], age [27–30], body mass [27, 31, 32], body height [27, 31, 33], dominant side [28, 34–36], and physical activity [37] impact isometric and isokinetic strength of muscles and consequently joints.

Apart from these anthropometric values and subject body characteristics, the amount of torque generated at a joint is limited by the number of muscles, tendons, and joint constraints. Force-velocity scaling, force-length scaling, nonlinear passive force production owing to ligaments and connective tissue around a joint, and muscle activation dynamics are among the fundamental muscular constraints. These biomechanical constraints can be either modeled with an anatomically-detailed muscle tension [38–40] or muscle torque generators (MTGs) [41]. Complicated wrapping geometry and redundancy should be taken into consideration when employing muscles [2, 3]. MTGs allow for the reduction of multibody system models by directly approximating force-length-velocity curves and passive components at the joint level (scaling torques using the joint kinematic biomechanical variable) [41]. We have adjusted the MTG models to consider anthropometric values and subject body characteristics.

So far, MTG models have only been implemented on 1-degree of freedom (DoF) joints [10, 41–43]. In 2018, McNally and McPhee [10] used a weighted approach to combine the shoulder adduction/abduction (SAA) with shoulder flexion/extension (SFE) torque [10, 44]. Recently, Bell and McPhee [45] evaluated a coupling effect of shoulder motions and developed a 2-DoF shoulder model. To date, multi-DoF joints, e.g., torso, neck, or hip, have not been comprehensively modeled by MTGs.

To model and simulate human MSK systems, several commercial (e.g., AnyBody, ADAMS-LifeMOD, MuJoCo, SIMM, SimMechanics, MapleSim) and open-source (e.g., OpenSim, MSMS, RBDL, BiomechZoo) multibody dynamics simulation software have been used [2]. Except for SimMechanics and MapleSim, it is challenging to use the above software for augmenting the human model with sports tools, active prostheses, or exoskeleton models [2], especially when using hydraulic, pneumatic, or flexible components such as bike tires [18] or golf shafts [44]. MapleSim is based on symbolic computing and provides analytic derivatives when necessary, such as for optimization or sensitivity analyses, whereas the other software mentioned are based on numerical computation.

In summary, this work's distinguishing characteristics and contributions are:

A. An adjustable, open-source, user-friendly, and easily augmentable (as an external tool) biomechanical MSK (MapleSim Biomechanics) model capable of extracting the model equations symbolically.
B. Modeling of 20-DoF upper body skeletal dynamic model that is adjustable with anthropometric values: (I) sex, (II) body mass, and (III) body height.
C. Modeling of biomechanical joint torques with adjusted MTGs according to anthropometric values and subject body characteristics: (I) sex, (II) age, (III) body mass, (IV) body height, (V) dominant side, and (VI) physical activity.
D. Multi-DoF joint implementation of biomechanical joint torques using an MTG model.
E. Providing a general model useful in various applications broadly categorized as motion analysis, sports engineering, design and control of biomechatronic devices.

First, this paper presents the locations and inertia parameters of the upper body model scaled by anthropometric parameters in sections 3.1 and 3.2. Second, the biomechanical joint torque model is introduced in section 4 as a function of biomechanical kinematic variables, anthropometric parameters, and subject body characteristics. Then, the symbolic model extraction is presented in section 5, two simulation methods of inverse and forward dynamics are introduced in section 6, and the model is validated with the literature data in section 7. The method of model simulation is described and an example is provided. Finally, applications of the provided model are discussed in section 8.

2. Anthropometric and subject characteristics

The biomechanical MSK model relies on anthropometric values and subject body characteristics. These variables that adjust the MSK MapleSim Biomechanics model consist of:

- **Sex**: impacts muscle morphology and control, with men having bigger muscles that produce more consistent and stronger contractions than females [46]. Consequently, males have higher joint isometric and isokinetic torques than females [27]. However, the sex relation does not follow a rule of thumb formula for all joints and each-joint comparison should be reported separately. Furthermore, sex variables impact the anatomical locations, the COMs, and inertia parameters [25, 26].
- **Age**: has a significant impact on an individual's physical capabilities. The age variable is negatively correlated with all strength measures in both sexes [28, 46]. The muscle peak isokinetic strength and power in adults (more than 20 years old) gradually declines as age increases. However, muscle strength decline rates differ in individual muscles and, consequently, joints [27, 29, 30].
- **Body height**: has substantial power function connections to isometric force and torque measurements and positively impacts joint strength [27, 31, 33]. Furthermore, the body height impacts the anatomical locations and segment COMs [25, 26].

- **Body mass**: has a high positive connection to muscle mass and is positively associated with joint torque [27, 31, 32]. Additionally, body mass impacts the inertia parameters [25, 26].
- **Dominance**: appears to be more widespread in the upper extremities with minor variances [28, 34–36, 46]. The dominant side can provide a mean value of 9% more strength than the non-dominant side [34, 36].
- **Physical activity**: has an impact on muscle strength [47]. Specifically, adult athletes have significantly higher concentric and eccentric peak strength of most muscles [37].

The six variables of (I) sex, (II) age, (III) body mass, (IV) body height, (V) dominant side, and (VI) physical activity are used to adjust the subject MSK model in sections 3 and 4.

3. Skeletal dynamic model

3.1. Locations of anatomical landmarks

This section presents the location of anatomical landmarks and segment COMs of the upper body using anthropometric data: subject sex, body height. The pelvis and the lower limb are fixed to the ground, and the global frame is attached to the pelvis anatomical coordinate system according to International Society of Biomechanics (ISB) standard [48]. Because no suitable joint center estimation for the Thoracic joint center is available, the torso (a single segment) stands for Thorax and Abdomen [25].

The orientations of the segment coordinate systems are created, and the joint centers are determined according to the literature using specified anatomical landmarks (Figure 1) [25, 48]. The definitions and abbreviations of the segment anatomical landmarks are adopted from [25, 48]. Using these critical landmarks, the segment anatomical coordinate systems are defined by [25, 26, 48] and presented in Table 1.

The locations of segment anatomical landmarks and the COMs shown for the right body side in Figure 1 are adopted from [25]. Then, the raw position data was scaled to mean body height for each sex (male mean stature 1.77 m, female mean stature 1.61 m) and are presented in Table 2.

The right and left limbs are considered to be symmetrical. Transforming the position of anatomical landmarks and segment COMs from the right limb (reported in Table 2) to the left limb is done by the matrix in Equation (1).

$$R_{R2L} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{1}$$

3.2. Body segment inertia parameters

The biomechanical modeling and analysis of human movement and posture rely heavily on body segment inertial parameterss (BSIPs) [49]. BSIPs are often calculated using linear or non-linear regression equations published in the literature [25, 26, 50, 51]. This section presents the body segment inertia parameter estimation by anthropometric data: subject sex, body height, and body mass.

Table 1.: Abbreviations and descriptions for anatomical coordinate systems used in the upper body kinematic model according to ISB standard [25, 26, 48].

Segment	Origin /Axis	Description (refer to Figure 1)
Pelvis	O	Coincident with the LJC
	Z	The line connecting the LASIS to the RASIS
	Y	The line perpendicular to the plane through the LASIS, the RASIS, and the MPSIS, pointing cranially
	X	The cross product of the Y- and Z-axes
Torso	O	Coincident with the CJC
	Y	The line connecting the LJC to the CJC, pointing superiorly
	Z	The line perpendicular to the plane containing the LJC, the CJC, and the SUP, pointing laterally
	X	the cross product of the Y- and Z-axes
Head	O	Coincident with the CJC
	Y	The line connecting the CJC to the HV, pointing superiorly
	Z	the line perpendicular to the plane containing the HV, the CJC, and the SEL, pointing laterally
	X	the cross product of the Y- and Z-axes
Upper arm	O	Coincident with the SJC
	Y	The line connecting the EJC to the SJC, pointing proximally
	X	The line perpendicular to the plane containing the SJC, the LHE, and the MHE, pointing anteriorly
	Z	The cross product of the X- and Y-axes
Forearm	O	Coincident with the EJC
	Y	The line is connecting the WJC and the EJC, pointing proximally
	X	The line perpendicular to the plane through the EJC, the US, and the RS, pointing anteriorly
	Z	The cross product of the X- and Y-axes
Hand	O	Coincident with the WJC
	Y	The line connecting the midpoint between the MH2 and the MH5 to the WJC, pointing proximally
	X	The line perpendicular to the plane containing the WJC, the MH2, and the MH5, pointing anteriorly
	Z	The cross product of the X- and Y-axes

Table 2.: Scaled 3D positions of anatomical landmarks and segment COMs relative to segment anatomical coordinate systems for female and male (right limb) [25, 26].

Segment	Reference Coordinate System	Anatomical Landmark (refer to Figure 1)	Scaled 3D position to height (%) Female X	Female Y	Female Z	Male X	Male Y	Male Z
Pelvis	LJC	LASIS (Left Antero Superior Iliac Spine)	5.40	-0.81	-7.39	4.41	0.40	-6.33
		RASIS (Right Antero Superior Iliac Spine)	5.40	-0.81	7.39	4.41	0.40	6.33
		MPSIS (Midpoint between the Postero-Superior Iliac Spines)	-6.71	-0.81	0.00	-5.76	0.40	0.00
Torso	CJC	C7 (7th Cervicale)	-3.54	2.11	0.00	-3.90	1.92	0.00
		SUP (Suprasternale)	2.80	-2.42	0.00	2.60	-2.54	0.00
		RA (Right Acromion)	-1.86	-1.37	11.24	-1.19	-1.75	11.81
		SJC (Shoulder Joint Centre)	0.81	-4.53	11.24	1.19	-4.12	11.81
		LJC (Lumbar Joint Centre)	0.00	-26.52	0.00	0.00	-27.01	0.00
		COM	-0.43	-11.55	-0.12	0.96	-11.36	-0.06
Head	CJC	HV (Head Vertex)	0.00	13.73	0.00	0.00	13.79	0.00
		SEL (Sellion)	4.91	7.89	0.00	4.69	7.74	0.00
		COM	-0.93	8.20	0.00	-0.85	7.68	0.00
Upper arm	SJC	RA (Right Acromion)	-2.17	3.48	0.25	-1.41	3.05	0.17
		LHE (Lateral Humeral Epicondyle)	0.00	-13.73	2.11	0.00	-14.58	2.32
		MHE (Medial Humeral Epicondyle)	0.00	-14.78	-2.11	0.00	-14.92	-2.32
		EJC (Elbow Joint Centre)	0.00	-14.25	0.00	0.00	-14.75	0.00
		COM	-0.99	-6.46	-0.37	0.28	-6.67	-0.40
Forearm	EJC	US (Ulnar Styloid)	0.00	-15.03	-1.68	0.00	-16.05	-1.86
		RS (Radial Styloid)	0.00	-15.65	1.68	0.00	-16.05	1.86
		WJC (Wrist Joint Centre)	0.00	-15.34	0.00	0.00	-16.05	0.00
		COM	0.31	-6.34	0.31	0.17	-6.67	0.23
Hand	WJC	MH2 (2nd Metacarpal Head)	0.00	-4.91	2.36	0.00	-4.86	2.60
		MH5 (5th Metacarpal Head)	0.00	-3.91	-2.36	0.00	-4.24	-2.60
		FT3 (3rd Fingertip)	0.87	-10.31	-0.81	0.40	-10.68	0.06
		COM	0.31	-3.42	0.19	0.40	-3.84	0.34

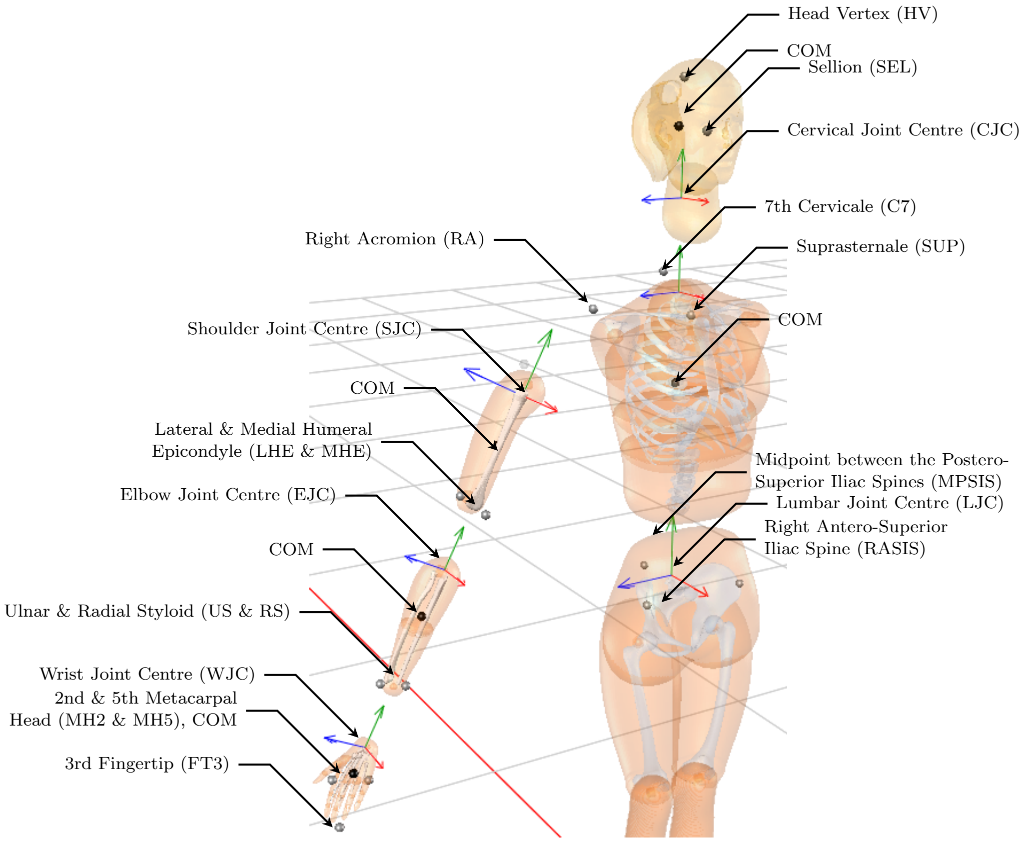


Figure 1.: Locations of selected anatomical landmarks and the consequent orientations of the segment coordinate system according to ISB standard [25, 48].

The mass of each segment (M), provided as a percentage of the total body mass, is presented in Table 3 [26]. The inertia tensor is calculated relative to the origin of segment coordinate axes. Explicitly, the segment inertia matrix (I) is expressed relative to the segment length (L) and the segment mass (M) with Equation (2) for the right limb. Scaled segment length L to body height, scaled segment mass (M) to body mass, and scaling factors r_{ij} for the inertia matrix are provided in Table 3.

$$I_R = ML^2 \begin{bmatrix} r_{xx}^2 & r_{xy}^2 & r_{xz}^2 \\ r_{xy}^2 & r_{yy}^2 & r_{yz}^2 \\ r_{xz}^2 & r_{yz}^2 & r_{zz}^2 \end{bmatrix} \tag{2}$$

The segment inertia matrix of the left limb can be calculated by transforming the corresponding right limb inertia matrix of Equation (2) by Equation (3) using the matrix of Equation (1).

$$I_L = R_{R2L}^T I_R R_{R2L} \tag{3}$$

Table 3.: The female and male segment scaled mass, scaled length, and scaling factors for the inertia tensor [25, 26]. i is the unit imaginary number.

Segment	Origin	Length definition	Scaled mass (%)		Scaled length (%)		Scaling factors for the inertia tensor $[r_{xx}, r_{yy}, r_{zz}, r_{xy}, r_{xz}, r_{yz}]$(%)	
			Female	Male	Female	Male	Female	Male
Torso	CJC	CJC to LJC	30.4	33.3	26.65	27	$[29, 27, 29, 22, 5, 5i]$	$[27, 25, 28, 18, 2, 4i]$
Head	CJC	CJC to HV	6.7	6.7	13.11	13.8	$[32, 27, 34, 6i, 1, 1i]$	$[31, 25, 33, 9i, 2i, 3]$
Upper arm	SJC	SJC to EJC	2.2	2.4	15.09	15.3	$[33, 17, 33, 3, 5i, 14]$	$[31, 14, 32, 6, 5, 2]$
Forearm	EJC	EJC to WJC	1.3	1.7	15.34	16	$[26, 14, 25, 10, 4, 13i]$	$[28, 11, 27, 3, 2, 8i]$
Hand	WJC	WJC to FT3	0.5	0.6	10.37	10.7	$[27, 18, 25, 12, 10, 12i]$	$[26, 16, 24, 9, 7, 8i]$

3.3. Degrees of freedom

The presented upper body kinematic model has 20 DoF: 3 DoF for the torso (torso flexion/extension (TFE), torso right/left lateral flexion (TRLLF), and torso right/left axial rotation (TRLAR)), 3 DoF for the neck (neck flexion/extension (NFE), neck right/left lateral flexion (NRLLF), and neck right/left axial rotation (NRLAR)), 3 DoF for each shoulder (SFE, SAA, and shoulder medial/lateral rotation (SMLR)), 2 DoF for each elbow (elbow flexion/extension (EFE) and forearm pronation/supination (FPS)), and 2 DoF for each wrist (wrist flexion/extension (WFE) and wrist radial/ulnar deviation (WRUD)). The definitions and movements of DoF in the upper body kinematic model are presented in Table 4 (visualized in Figure 2). The left limb has reversed directions for WRUD, FPS, SAA, and SMLR compared to Table 4 (right limb information).

4. Biomechanical joint torque model

Human muscles have several recognized constraints that limit the amount of torque generated around each joint. The critical muscle's constraints can be listed as follows: force-velocity scaling, force-length scaling, nonlinear passive force generation due to ligaments and connective tissues surrounding a joint, and muscle activation dynamics.

Generally, a 3-element Hill-type muscle-tendon unit (Figure 3a) is used for muscle modeling [38, 39]. Furthermore, moving and wrapping points are used to simulate the constraints of the muscle-tendon geometry [39, 40]. Each muscle in Figure 4 is divided into two parts: an active contractile portion (creates active force) and a passive parallel-elastic element (with viscoelasticity). Moreover, a nonlinear series-elastic element describing the tendon's mechanical characteristics completes the model. These type of complex muscle models are difficult to obtain. Moreover, the model variables cannot be experimentally verified as they need in vivo human studies.

MTG model offers a joint torque that imitates the performance of muscles and joint constraints without the need for modeling muscle geometries and forces (Figure 3b) [41]. Explicitly, the MTG model mimics elements of muscle modeling: the velocity constraint and the joint angle constraint [10, 41] (shown in Figure 5). The MTG models, in torque-driven simulation, deliver torque at every joint that is a function of that joint's kinematics (monoarticular assumption) [68].

Each MTG model has a unique set of characteristics: an active component (generates the active torque limited by the joint's angle and angular velocity) and a passive component (mimicking the joint's small viscosity or friction and constraining the joint ROM). The human joint torque (τ_h) calculated by the MTG model is shown in Equation (4) [10, 41].

$$\tau_h\left(t\right) = \tau^{+}_{u2a}\left(u^{+}, t\right)\tau^{+}_{\omega}\left(\omega\right)\tau^{+}_{\theta}\left(\theta\right)\tau^{+}_{0} + \tau^{-}_{u2a}\left(u^{-}, t\right)\tau^{-}_{\omega}\left(\omega\right)\tau^{-}_{\theta}\left(\theta\right)\tau^{-}_{0} + \tau_p\left(\theta, \omega\right) \quad (4)$$

where

$\tau_{u2a}\left(u, t\right)$ the excitation-to-activation signal ordinary differential equation (ODE) function, which is introduced in section 4.5 and Equation (9)

$\tau_{\omega}\left(\omega\right)$ the active-torque-angular-speed-scaling function, which is introduced in section 4.4 and Equation (8)

$\tau_\theta(\theta)$	the active-torque-position-scaling function, which is introduced in section 4.3 and Equation (7)
$\tau_p(\theta,\omega)$	the passive torque function due to viscous damping and nonlinear stiffness, which is introduced in section 4.1 and Equation (5)
τ_0	the peak isometric joint strength producing capability

The product of $\tau_{u2a}(u,t)\,\tau_\omega(\omega)\,\tau_\theta(\theta)\,\tau_0$ is the net active joint torque that the biomechanical model generates. The control input to the MTG model is the activation signal $a(t)$ or the excitation signal $\varepsilon(t)$, which may take any value between 0 and 1. The examples of these four functions are shown in Figure 6.

Two active components of MTGs (42 total products of $\tau_{u2a}(u,t)\,\tau_\omega(\omega)\,\tau_\theta(\theta)\,\tau_0$) were used to control each DoF (joint axis of rotation): one for each direction (negative and positive) of joint rotation (two for each joint). One passive component of MTGs (20 total $\tau_p(\theta,\omega)$) were used for each DoF (joint axis of rotation): one for each joint. Consequently, the MSK system is over-actuated and requires actuation redundancy resolution, potentially with an optimization solution. The maximum torque range of each active component of MTGs was limited to τ_0^+ for positive MTGs and τ_0^- for negative MTGs.

The parameters of Equations (4-8) can be calculated by fitting the functions to the experimental data. Accordingly, a computerized robotic dynamometer is generally utilized in sports and orthopedic medical applications for acquiring experimental data. The human dynamometer is a device that measures the net torque generated by an isolated joint in the body. It can be used in both isometric and isokinetic joint situations.

4.1. Passive torque function

Passive torques arise when the muscles, tendons, and ligaments surrounding the joints are strained [69]. Passive joint torques are intensified near the anatomical joint limits [69–71]. The double exponential function in Equation (5) is typically used to simulate the joint's viscous damping and nonlinear stiffness [72]. Consequently, this function limits the joint position within its ROM [72]. Specifically, when θ falls outside the ROM (those defined by breakpoints θ^-_{max} and θ^+_{max}), it creates a high restoring torque. In addition, the rotational damping term is included in Equation (5) to represent viscoelasticity.

$$\tau_p(\theta,\omega) = k_1 e^{-k_2(\theta-\theta^-_{max})} - k_3 e^{k_4(\theta-\theta^+_{max})} - c\omega \tag{5}$$

where

θ	the joint angle
k	the passive parameters determined by fitting the nonlinear function to experimental data
θ^-_{max}	the maximum ROM for negative direction, which is presented in Table 4
θ^+_{max}	the maximum ROM for positive direction, which is presented in Table 4
c	the rotational damping linear coefficient

The passive parameters (k_i) are generally determined by fitting the nonlinear function to experimental data and are between 0.5 to 6 [10]. k_i should be lower for the

Table 4.: Abbreviations and definitions of DoF, the mean range of motion (ROM) for female and male, the mean peak angular velocity of the upper-body joints, and the polynomial coefficients of torque-angle scaling functions (normalized isometric torque functions).

Joint	DoF	Axis		Direction	θ_{max} (rad) Female	θ_{max} (rad) Male	Ref	$\lvert\omega_{max}\rvert$ (rad/s) Mean	Ref	Polynomial Coefficient of Torque-angle Scaling Function in Equation (7) p_1	p_2	p_3	Ref
Torso	TFE	Z-axis	-	Flexion	0.547	0.620	[52]	2.531	[53]	0.9466	0.3969	-0.7370	[54]
			+	Extension	-0.470	-0.469	[52]	3.421	[55]	0.6093	0.7892	-0.2903	[54]
	TRLLF	X-axis	+	Right Lateral Flexion	0.354	0.412	[52]	1.344	[56]	0.9995	0.0485	-1.0856	[24]
			-	Left Lateral Flexion	-0.354	-0.412	[52]	1.152	[56]	0.9671	-0.4460	-1.5055	[24]
	TRLAR	Y-axis	-	Right Axial Rotation	-0.730	-0.868	[52]	3.857	[56]	0.9903	0.1555	-0.6199	[24]
			+	Left Axial Rotation	0.730	0.868	[52]	3.176	[56]	0.9740	-0.2235	-0.4801	[24]
Neck	NFE	Z-axis	-	Flexion	0.972	1.004	[57]	1.134	[58]	0.7678	0.2558	-0.0337	[31]
			+	Extension	-1.288	-1.162	[57]	1.117	[58]	0.9834	0.0974	-0.1432	[31]
	NRLLF	X-axis	+	Right Lateral Flexion	0.753	0.680	[57]	0.384	[59]	0.9996	0.0266	-0.3572	
			-	Left Lateral Flexion	-0.756	-0.770	[57]	0.384	[59]	0.9667	-0.2524	-0.4777	
	NRLAR	Y-axis	-	Right Axial Rotation	-1.360	-1.326	[57]	0.541	[59]	0.9904	0.0492	-0.0630	
			+	Left Axial Rotation	1.331	1.252	[57]	0.541	[59]	0.9737	-0.0729	-0.0506	
Shoulder	SFE	Z-axis	+	Flexion	2.965	2.904	[52]	4.974	[56]	0.9992	0.0131	-0.0565	
			-	Extension	-1.124	-1.092	[52]	3.491	[56]	0.6621	0.3796	-0.1066	
	SVAA	X-axis	+	Vertical Adduction	0.199	0.283	[60]	4.084	[56]	0.8255	-0.3164	-0.1433	
			-	Vertical Abduction	-2.985	-2.989	[52]	4.538	[56]	0.8298	0.5322	0.1398	
	SHAA		+	Horizontal Adduction	1.018	0.948	[52]	4.272		0.9253	0.0919	-0.0182	
			-	Horizontal Abduction	-2.393	-2.281	[52]	4.274		0.8988	-0.2512	-0.1557	
	SMLR	Y-axis	+	Medial Rotation	1.120	0.971	[52]	8.954	[56]	0.9390	-0.2040	-0.1702	[61]
			-	Lateral Rotation	-1.571	-1.521	[52]	5.114	[56]	0.9676	-0.1777	-0.2430	[61]
Elbow	EFE	Z-axis	+	Flexion	2.402	2.344	[52]	6.231	[55]	0.4046	0.9405	-0.3714	[62]
			-	Extension	0.093	0.101	[52]	4.451	[56]	0.4932	0.8047	-0.3194	[62]
	FPS	Y-axis	+	Pronation	2.859	2.774	[63]	5.184	[56]	0.7748	0.3948	-0.1731	[64]
			-	Supination	-0.010	0.083	[63]	7.121	[56]	0.3414	0.7590	-0.2187	[64]
Wrist	WFE	Z-axis	+	Flexion	1.489	1.446	[52]	9.913	[65]	0.9996	0.0228	-0.3021	[62]
			-	Extension	-1.391	-1.313	[52]	8.029	[56]	0.9860	0.1331	-0.3139	[62]
	WRUD	X-axis	+	Ulnar Deviation	0.281	0.284	[66]	2.740	[56]	0.9728	-0.4350	-1.7343	[67]
			-	Radial Deviation	-0.401	-0.382	[66]	3.683	[56]	0.9608	-0.5288	-1.7812	[64]

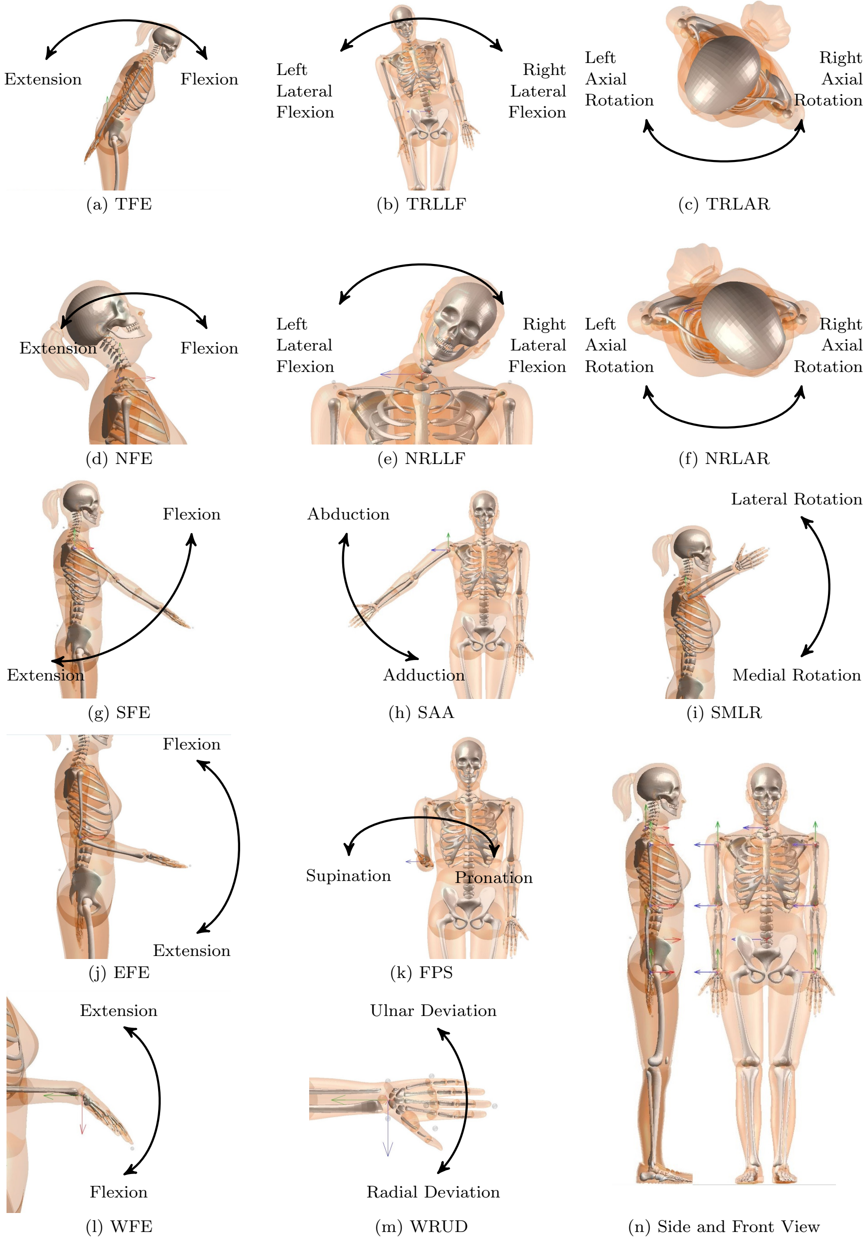
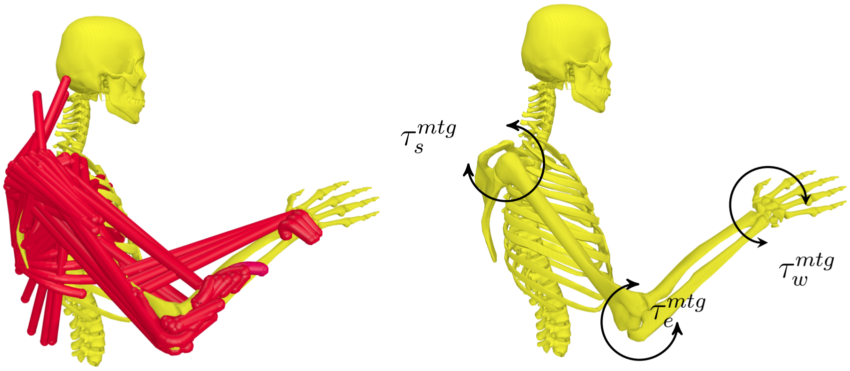


Figure 2.: Depiction of MSK model DoF and directions for (a) TFE, (b) TRLLF, (c) TRLAR, (d) NFE, (e) NRLLF, (f) NRLAR, (g) SFE, (h) SAA, (i) SMLR, (j) EFE, (k) FPS, (l) WFE, and (m) WRUD. (n) Side and Front view with the initial posture (zero joint angles).

weaker joints like NRLLF, NRLAR, FPS, WFE, and WRUD, based on our experimental experience. The linear damping coefficient (c) is recommended to be about 0.1 Nm/rad [72].

In Table 4, θ_{max} for the WRUD, FPS, shoulder vertical adduction/abduction (SVAA), and SMLR DoF of the left limb is negative.



(a) Anatomically detailed muscles (b) Muscle torque generators

Figure 3.: Schematic of the shoulder, elbow, and wrist joint, actuated by (a) Hill-type muscles and (b) MTGs. Note the complex muscle route geometry and redundancy encountered when using muscles.

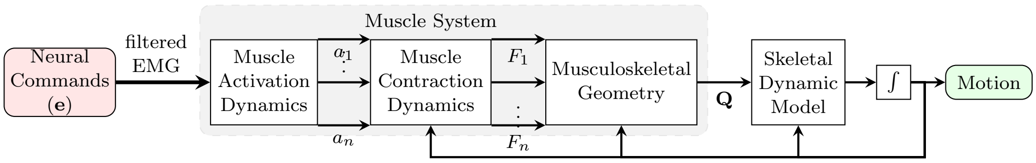


Figure 4.: Forward simulation of a musculoskeletal system with multiple muscles. a_i and F_i are the activation signal and the muscle force in the dynamic muscle model, respectively.

4.2. Peak joint strength

Mean peak isokinetic torque for non-athletic females and males with their related anthropometric measures are presented in Table 8. The mean maximal isometric joint torque for any non-athletic adult male or female is estimated from parameters in Table 8. The intercept and sex parameters in Table 8 are extrapolated based on the anthropometric parameters and specific subjects reported in Table 8. The mean peak isokinetic torque for concentric contraction is regressed with Equation (6).

$$\tau_0 = (\beta_1 + \beta_2 \times s + \beta_3 \times a + \beta_4 \times h + \beta_5 \times m) \times \left(1 - \frac{\mu}{2} + \mu \times d\right) \times (1 + \varsigma \times w) \quad (6)$$

where

β_i	the torque-anthropometric parameters reported in Table 8
s	the sex parameter (for a male is 1 and for a female is 0)
a	the age of the subject in year
h	the height of the subject in meter
m	the body mass of the subject in kilogram
w	the physical activity or workout sessions per week
d	the dominant side (for a dominant side is 1 and for a non-dominant side is 0)
μ	the maximum strength difference of dominant with non-dominant side

ς	the effect of workout on peak muscle strength

In Table 8, the joint torque is reported for the average of both sides. The dominant side can provide a mean value of 9% more strength than the non-dominant side [34, 36]. Thus, the side parameter (μ) is selected as 0.09 for the arms. The central joints (torso and neck) are assumed to be symmetric in terms of maximum strength with $\mu = 0$. In Table 8, the joint torque is reported for healthy non-athletic subjects (w=0 for this group). For the upper-limb, the value of ς is equal to 0.065, which is a mean value reported by [47] for training time per week.

All the data reported in Table 8 are relevant to joint isometric torque except for the Horizontal Adduction and Abduction. Since the maximum isometric data are unavailable, the maximum isokinetic torque is reported.

4.3. Active-torque-angle scaling function

The tendon force-length relation is not linear, and accordingly, the joint isometric torque is not consistent in the full ROM. The active-torque-angle relationship is typically represented in the literature using second-order or fourth-order polynomial functions [42, 43, 78, 79] with the summation notation in Equation (7).

$$\tau_\theta\left(\theta\right) = ramp\left(\sum_{k=1}^{n+1} p_k \theta^{k-1}\right) \tag{7}$$

where

θ	the joint angle
p_k	the polynomial coefficient, which is presented in Table 4
n	the order of polynomial functions

and $\tau_\theta\left(\theta\right)$ is normalized by the maximal isometric torque of each joint. Consequently, the value of $\tau_\theta\left(\theta\right)$ at the optimum angle is 1.

For fitting the polynomial functions to the experimental data, the maximum isometric torque should be available for all ROMs. Since there is not enough published information for all ROMs, for example, the shoulder joint [62], the fitted polynomial function may go to zero or even negative for ranges out of the reported angles. Thus, we introduce two solutions. The first solution is using a dynamometer to measure the joint isometric torque for all ROMs that we use for shoulder joint (Flexion, Extension, Vertical Adduction, Vertical Abduction, Horizontal Adduction, Horizontal Abduction) for one male subject (25 years, 75 kg mass, and 1.71 m height) with a System 4 Dynamometer from Biodex Medical Systems (Mirion Technologies, Inc., NY, USA). The second solution is introducing "edge torque". The edge torque is the estimated or assumed joint torque at the maximum joint angle, which is calculated with the average of fitted polynomial and interpolation with nearest joint torque.

As mentioned, the fitted polynomial function must have a maximum value of 1. For the normalization, in contrast to [41], which used the maximum of joint torque, we use the maximum of the fitted polynomial function.

For the neck joint (NRLLF and NRLAR), no experimental data is available. Thus, we have used the torso-relevant data with scaling the torso's ROM to match the neck's ROM.

Table 8.: Peak joint torque for non-athletic females and males (mean of dominant and non-dominant limb) and estimated parameters of peak isokinetic torque for non-athletic subjects with the anthropometric measures: body age, height, mass, and sex.

Joint	DoF	Direction	Female						Male						β_1 (Nm)	β_2 (Nm)	β_3 (Nm/year)			β_4 (N)			β_5 (Nm/kg)		
			Torque (Nm)	Age (year)	Height (m)	Mass (kg)	Subject Numbers	Ref	Torque (Nm)	Age (year)	Height (m)	Mass (kg)	Subject Numbers	Ref			Mean	Range	Ref	Mean	Range	Ref	Mean	Range	Ref
Torso	TFE	-	109	30	1.65	60	43	[24]	162	31	1.75	75	59	[24]	-2.05	14.85	-1.6	-2.00 : -1.20	[30]	0.03	0.01 : 0.05	[33]	2.65	1.6 : 3.7	[32]
		+	189	30	1.65	60	43	[24]	288	31	1.75	75	59	[24]	77.95	60.85									
	TRLLF	+	101	30	1.65	60	43	[24]	141	31	1.75	75	59	[24]	-10.05	1.85									
		-	100	30	1.65	60	43	[24]	137	31	1.75	75	59	[24]	-11.05	-1.15									
	TRLAR	-	38	30	1.65	60	43	[24]	67	31	1.75	75	59	[24]	-73.05	-9.15									
		+	35	30	1.65	60	43	[24]	66	31	1.75	75	59	[24]	-76.05	-7.15									
Neck	NFE	-	15	30	1.64	65	5	[73]	30	32	1.77	77	11	[73]	22.41	13.95	-0.70	-1.05 : -0.34	[29]	0.36	0.20 : 0.53	[31]	0.2	0.06 : 0.33	[31]
		+	21	30	1.64	65	5	[73]	52	32	1.77	77	11	[73]	28.41	29.95									
	NRLLF	+	16	30	1.64	65	5	[73]	36	32	1.77	77	11	[73]	23.41	18.95									
		-	16	30	1.64	65	5	[73]	36	32	1.77	77	11	[73]	23.41	18.95									
	NRLAR	-	6	30	1.64	65	5	[73]	15	32	1.77	77	11	[73]	13.41	7.95									
		+	6	30	1.64	65	5	[73]	15	32	1.77	77	11	[73]	13.41	7.95									
Shoulder	SFE	+	34	25	1.62	62	12	[46]	61	25	1.72	75	12	[46]	-100.95	14.83	-0.17	-0.33 : -0.02	[27]	71	16 : 125	[27]	0.39	0.05 : 0.73	[27]
		-	45	25	1.62	62	12	[46]	95	25	1.72	75	12	[46]	-89.95	37.83									
	SVAA	+	43	25	1.62	62	12	[46]	84	25	1.72	75	12	[46]	-91.95	28.83									
		-	23	25	1.62	62	12	[46]	46	25	1.72	75	12	[46]	-111.95	10.83									
	SHAA	+	35	18	1.62	52	13	[74]	57	18	1.72	69	25	[75]	-97.24	8.27									
		-	31	18	1.62	52	13	[74]	52	18	1.72	69	25	[75]	-101.24	7.27									
	SMLR	+	27	25	1.62	62	12	[46]	51	25	1.72	75	12	[46]	-107.95	11.83									
		-	19	25	1.62	62	12	[46]	34	25	1.72	75	12	[46]	-115.95	2.83									
Elbow	EFE	+	33	23	1.66	67	24	[76]	63	24	1.80	81	30	[76]	-21.67	23.02	-0.12	-0.24 : -0.01	[27]	23.7	-6.5 : 54	[27]	0.27	0.09 : 0.46	[27]
		-	29	23	1.66	67	24	[76]	53	24	1.80	81	30	[76]	-25.67	17.02									
	FPS	+	6.3	28	1.68	62	22	[77]	12	30	1.78	80	21	[77]	-46.90	-1.29									
		-	5.8	28	1.68	62	22	[77]	11	30	1.78	80	21	[77]	-47.40	-1.79									
Wrist	WFE	+	15	25	1.68	59	78	[27]	25	24	1.79	74	90	[27]	-23.07	6.85	-0.04	-0.09 : 0.02	[27]	21.5	2.8 : 40.3	[27]	0.05	-0.08 : 0.18	[27]
		-	7	22	1.70	60	26	[35]	11	22	1.80	76	6	[35]	-31.67	1.05									
	WRUD	+	7	22	1.70	60	26	[35]	11	22	1.80	76	6	[35]	-31.67	1.05									
		-	8	22	1.70	60	26	[35]	12	22	1.80	76	6	[35]	-30.67	1.05									

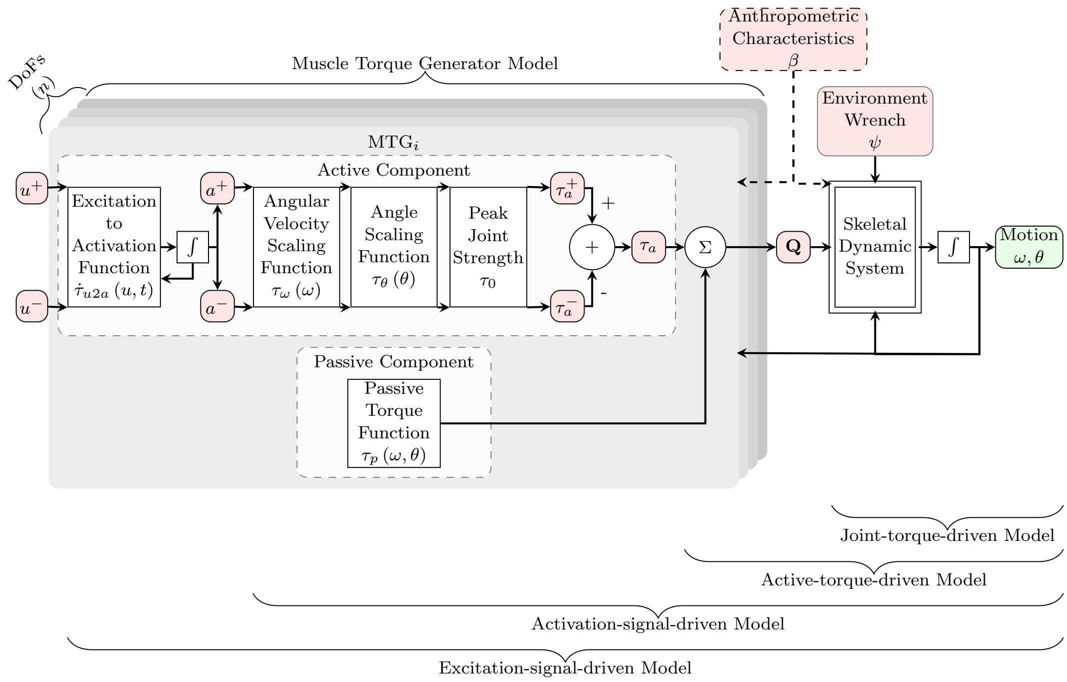


Figure 5.: Schematics of forward simulation of a musculoskeletal system with MTGs. The MTG model consists of 2 active components for two excitation signals (positive and negative direction) and one passive component.

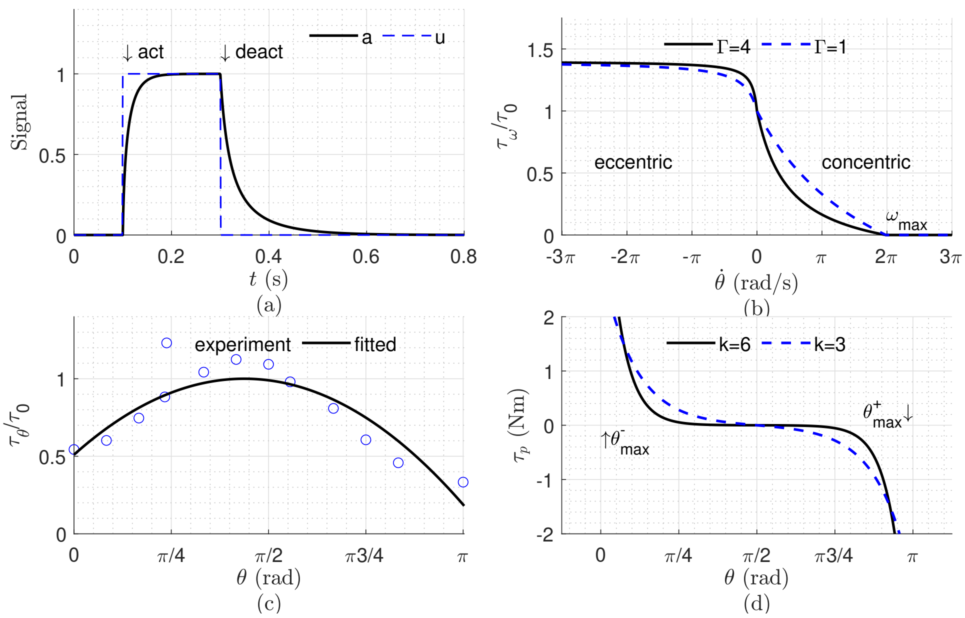


Figure 6.: Example of MTG curves to mimic constraints of tendon and muscle forces: activation signal curve (a), the active-torque-angular-speed scaling curve (b), the active-torque-angle scaling curve (c), and the passive torque curve (d).

The left limb has reverse positive and negative directions for WRUD, FPS, SVAA, and SMLR. In Table 4, the even index of p_k for the WRUD, FPS, SVAA, and SMLR

DoF of the left limb has a reversed sign.

4.4. Active-torque-angular-speed scaling function

Muscle strength is a function of muscle velocity. As a result, the angular velocity (ω) of the movement scales the activation torque (τ_a) (active-torqueangular-speed correlation). A rational piecewise instantaneous isokinetic torque function, associated with human muscle and predicted from a linearized Hill model structure, is implemented to represent this behavior according to [80–83]:

$$\tau_\omega(\omega) = \begin{cases} 0 & \omega > \omega_{max} \\ \frac{|\omega_{max}|-\omega}{|\omega_{max}|+\Gamma\omega} & \omega_{max} \geq \omega \geq 0 \\ \frac{(1-R)|\omega_{max}|+S\omega R(\Gamma+1)}{(1-R)|\omega_{max}|+S\omega(\Gamma+1)} & \omega < 0 \end{cases} \tag{8}$$

where

ω	the joint's instantaneous angular velocity
$\|\omega_{max}\|$	the absolute value of joint's peak angular velocity corresponding to zero torque production, as reported in Table 4
Γ	the shape factor influencing the curvature of the torque-velocity hyperbola concentric relationship
S	the slope of eccentric and concentric functions at zero angular velocity (the transition of eccentric to concentric phases) defined by [82]
R	the ratio of the maximum eccentric isokinetic torque (τ_m) over the maximum isometric torque (τ_0)

In Equation (8), for the $\omega \geq 0$ condition, $\tau_\omega(\omega)$ has a potential singularity as ω proceeds toward $-|\omega_{max}|/\Gamma$ instantaneous angular velocity. Consequently, the force-velocity scaling relationship differs from Hill-type's model [39] when the force is greater than the isometric tension (τ_0); it also has different output when $\omega < 0$ [82, 84]. As a result of $\tau_\omega(\omega)$ normalization by the maximal isometric torque of each joint, the value of $\tau_\omega(\omega)$ is 1 at zero velocity.

The value for Γ is about 3 [44]. The value for S is about 2 [10, 82] to 3 [85]. In addition, the value for R is around 1.4 [81] to 1.5 [82]. For negative direction movements (e.g., flexion, adduction, or supination), negative ω was used in Equation (8).

The absolute value of the joint's peak angular velocity for the shoulder horizontal abduction/adduction is not available; we have used the mean value of shoulder flexion/extension and vertical abduction/adduction instead.

4.5. Excitation-to-activation signal function

The muscle fiber recruitment status is represented by an excitation-to-activation signal function or simply activation dynamics [86]. The activation dynamics introduce a time delay between input and output, or the time between receiving a signal (neural excitation) and converting those signals into action potentials to activate muscle fibers by electrochemical processes in the muscle tissue [86]. Post-processed electromyography (EMG) data from experiments can be used to generate a collection of activations [87].

Similarly, for muscle groups around a joint, the activation signal generated by a neural drive is mimicked utilizing an extension of the activation curve [85] (independent of joint angle). Theoretically, the activation signal function incorporates muscle activation and deactivation time when the torque generator engages and disengages, respectively. Practically, this excitation-to-activation signal function is analogous to the temporal lag caused by the electrochemical activity generated by the action potential that leads to muscle contraction.

The modified function of an excitation-to-activation signal is suggested in Equation (9) adopted from the muscle fiber Hill-type model [88]. The excitation-to-activation dynamic model consists of a linear first-order differential equation and a non-linear function [89, 90]. The first-order differential equation converts the excitation signal $u\,(t)$ (from the CNS) to activation signal $a\,(t)$ which indicates the muscle excitation strength level.

$$\dot{\tau}_{u2a} = \begin{cases} \frac{u-a}{t_{act}(0.5+1.5a)} & e > a \\ \frac{(u-a)(0.5+1.5a)}{t_{deact}} & e \leq a \end{cases} \tag{9}$$

where

u	the excitation signal
a	the activation signal
t_{act}	the muscle activation time constants
t_{deact}	the muscle deactivation time constants

t_{act} can have a value of $40 + 0.25 \times$ age ms for an adult subject [91] and t_{deact} is assumed to be equal to $4t_{act}$. The excitation signal is in the range of $[0, 1]$ for monoarticular muscles. However, even for monoarticular muscles, the antagonist and agonist muscles activate simultaneously. In addition, the biarticular muscle provides tension for two joints [92, 93]. However, the MTG models assume a monoarticular joint. To overcome this assumption, the impact of the excitation signal of one joint on another joint should be investigated [45]. Studying the excitation signal of joints is out of the scope of this paper.

4.6. Multi-DoF joint implementation

Transferring these MTG torques to human joint torque depends on the joint type: single-DoF and multi-DoF. Most of human joints have consistent parameters in different orientations, denoted as one-DoF MTGs, e.g., torso, neck, forearm, and wrist. Consequently, the modeled MTGs are not a function of the coordinates for other-DoF. The MTG model for a single-DoF joint can directly apply to the human joint torque [43, 94].

A few human joints have variable parameters in different orientations, for example, the shoulder and hip joint (except for the axial rotation since it is configuration-independent). Specifically, the MTG parameters of multi-DoF MTGs are based on the orientation of other-DoF (Figure 7). For example, there are different MTG variables for

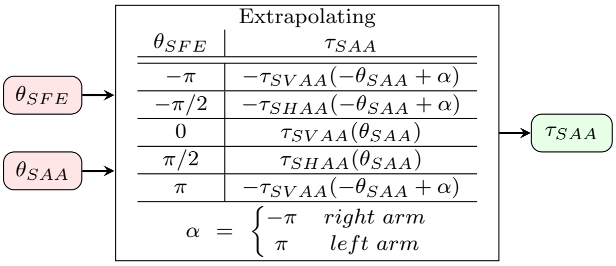


Figure 7.: The schematic for extrapolating the joint torque for x-direction rotation (SAA) as a function of the z-direction orientation (SFE). The minus sign of torque in the second column of the extrapolation table is neglected for normalized functions, for example, active-torque-angle and active-torque-angular-speed scaling functions.

the shoulder joint for abduction/adduction movement based on the value of the flexion/extension. Note that this joint orientation is defined with ZXY Euler angles. More specifically, when the upper arm has 0^o SFE in the z-direction, then the x-direction follows the MTG for SVAA. When the upper arm has 90^o SFE in the z-direction, then the rotated new x-direction follows the MTG for shoulder horizontal adduction/abduction (SHAA). The shoulder x-direction MTG elements are calculated with the extrapolation of MTG elements of SVAA and SHAA. Specifically, the x-direction shoulder MTG elements are computed with an extrapolation of SVAA (for 0^o SFE in the z-direction), SHAA (for 90^o SFE in the z-direction), and the shifted and mirrored SVAA (for 180^o SFE in the z-direction). MTG elements are presented in Table 4 and Table 8. The extrapolated elements are the final values of $\tau_\omega^{-/+}(\omega)$, $\tau_\theta^{-/+}(\theta)$, and $\tau_0^{-/+}$ for both positive and negative directions as well as $\tau_p(\theta, \omega)$ functions. Based on our tests, the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) method provides good curve fitting to experimental results in section 7.3.

5. General structure of dynamic equations

The dynamic musculoskeletal model can be exported as (I) joint-torque-driven, (II) active-torque-driven, (III) activation-signal-driven, and (IV) excitation-signal-driven, as shown in Figure 5. The joint torque-driven level consists of only the dynamic model of the skeletal system. The active-torque-driven level consists of the skeletal system and the passive torque function. The activation-driven level consists of the dynamic model of the skeletal system and the MTG model without the excitation-to-activation dynamic model. The forth form of the model consists of the skeletal system and full MTG model with the excitations as input.

5.1. Joint-torque-driven model

The unconstrained dynamic equations governing the MSK system response are a set of ODEs since the chosen n generalized coordinates and generalized speeds are independent. By utilizing the "Multibody Dynamic Exports" and "Optimized Code Generation" module of MapleSim, the equations of motion are converted to simplified C code that can be used in the MATLAB/Simulink environment [95, 96].

The joint-torque-driven (MTG-excluded) dynamic model is in the form of Equation (10) with three matrices comprising the system ODEs.

$$\mathbf{M}_{n\times n}\left(\boldsymbol{\theta}_{n\times 1}\right)\dot{\boldsymbol{\omega}}_{n\times 1}=\mathbf{F}_{n\times 1}\left(\boldsymbol{\omega}_{n\times 1},\boldsymbol{\theta}_{n\times 1}\right)+\mathbf{Q}_{n\times 1} \tag{10}$$

where

n	the number of independent coordinates and speeds
$\boldsymbol{\theta}$	the column matrix of all joint angles
$\boldsymbol{\omega}$	the column matrix of all joint angular speeds
$\mathbf{M}$	the mass matrix
$\mathbf{F}$	the right-hand side of the dynamic equations, which consist of Coriolis, centrifugal, and gravitational effects
$\mathbf{Q}$	the applied joint torques, a column matrix containing $\tau_h\left(t\right)$ for all joints

5.2. Active-torque-driven model

The active-torque-driven dynamic model is similar to the form of Equation (10) with the inclusion of the passive torque function in section 4.1. The active torques are converted to human joint torques using Equation (11). The activation-driven model combines Equations (10, 11).

$$\mathbf{Q}_{n\times 1}=\left[\tau_a+\tau_p\left(\boldsymbol{\theta},\boldsymbol{\omega}\right)\right]_{n\times 1} \tag{11}$$

5.3. Activation-signal-driven model

The activation-signal-driven (MTG-included) dynamic model is in the form of Equation (10) with three matrices comprising the system ODEs. The activation signals of positive and negative directions are converted to human joint torques using Equation (12), which is derived from Equation (4). The activation-driven model combines Equations (10,12).

$$\mathbf{Q}_{n\times 1}=\left[\boldsymbol{a}^{+}\boldsymbol{\tau}_{\omega}^{+}\left(\boldsymbol{\omega}\right)\boldsymbol{\tau}_{\theta}^{+}\left(\boldsymbol{\theta}\right)\boldsymbol{\tau}_{0}^{+}+\boldsymbol{a}^{-}\boldsymbol{\tau}_{\omega}^{-}\left(\boldsymbol{\omega}\right)\boldsymbol{\tau}_{\theta}^{-}\left(\boldsymbol{\theta}\right)\boldsymbol{\tau}_{0}^{-}+\boldsymbol{\tau}_{p}\left(\boldsymbol{\theta},\boldsymbol{\omega}\right)\right]_{n\times 1} \tag{12}$$

where

$^{+}$	the positive direction of joint
$^{-}$	the negative direction of the joint

5.4. Excitation-signal-driven model

Including the first-order differential equation of excitation-to-activation, established in Equation (9), introduces another state to the system as shown in Equation (13). Some researchers bypass this extra state with an alternative formulation (e.g., exponential gain formula), but their alternative approach does not consider the delay from excitation to activation [88].

$$\begin{bmatrix} I_{2n\times 2n} & 0 \\ 0 & \mathbf{M}_{n\times n}\left(\boldsymbol{\theta}_{n\times 1}\right) \end{bmatrix} \begin{bmatrix} \dot{\boldsymbol{a}}_{2n\times 1} \\ \dot{\boldsymbol{\omega}}_{n\times 1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\tau}_{u2a}\left(\boldsymbol{a}_{2n\times 1}, \boldsymbol{u}_{2n\times 1}, t\right)_{2n\times 1} \\ \mathbf{F}_{n\times 1}\left(\boldsymbol{\omega}_{n\times 1}, \boldsymbol{\theta}_{n\times 1}\right) + \mathbf{Q}_{n\times 1} \end{bmatrix} \tag{13}$$

$$\mathbf{Q}_{n\times 1} = \left[\boldsymbol{a}^{+}\boldsymbol{\tau}_{\omega}^{+}\left(\boldsymbol{\omega}\right)\boldsymbol{\tau}_{\theta}^{+}\left(\boldsymbol{\theta}\right)\boldsymbol{\tau}_{0}^{+} + \boldsymbol{a}^{-}\boldsymbol{\tau}_{\omega}^{-}\left(\boldsymbol{\omega}\right)\boldsymbol{\tau}_{\theta}^{-}\left(\boldsymbol{\theta}\right)\boldsymbol{\tau}_{0}^{-} + \boldsymbol{\tau}_{p}\left(\boldsymbol{\theta}, \boldsymbol{\omega}\right)\right]_{n\times 1} \tag{14}$$

6. Simulation method

6.1. Inverse dynamic simulation

Inverse dynamics involves solving the torque-driven equations of motion (provided in section 5.1) using the kinematics and the body characteristics to compute the resultant forces and moments at specific joints [97].

The MTG models have different positive and negative torques for one joint. This over-actuation requires actuation redundancy resolution, primarily by optimization methods. Variations of two optimization techniques are used to compute positive and negative active MTG torques: static and dynamic optimization. Static optimization (SO) is considered as an inverse approach: MTG torques are optimized in SO using time-marching, which means that time frames are solved consecutively [11]. Once the positive and negative active MTG torques are calculated, the inverse dynamic simulation can compute the positive and negative MTG activation signals.

6.1.1. Example

The resultant torques and signals are evaluated for a given motion, as an example of model application. The pick and place motions depicted in Figure 8 were measured from one subject (Male, 29-years, 1.80 m, 85 kg, right-handed, 1 workout sessions per week). The object weight is 40 N, and we assume the weight is equally spread on both palms of hands. The raw measured joint angles are filtered with a 5 Hz low pass filter (Figure 9). Using inverse dynamic simulation (given joint angles, velocities, and accelerations), the required torques for 20 joints are calculated using equation (10). The activation torques for 40 MTGs are derived using an optimization method and plotted in Figure 10. The activation torque is minimized to compute the joint torques using the "fmincon" function of MATLAB (R2021b. MathWorks, Natick, MA, USA). The activation and excitation signals, from Equation (4) are shown in Figure 11 and Figure 12.

According to the results in Figure 11, SFEs for both arms activate with 50% of the signal, and the TFE activates with roughly 100% when the object is lifted from the furthest target. This activation pattern means that lifting in the pose shown in Figure 8 has an undesirable impact on the torso and shoulder joint.

6.2. Forward dynamic simulation

The human CNS, which includes the brain and the spinal cord, controls the human body's mobility [98, 99]. Despite uncertain/unknown trajectories and intricate muscle dynamics, the CNS simultaneously maintains the kinetics and kinematics [99, 100]. Experimental investigations of human mobility suggest that the CNS optimally coordinates body movements [2, 101–103]. According to many studies, the CNS regulates the human by minimizing a cost function that may include jerk [104], torque [105],

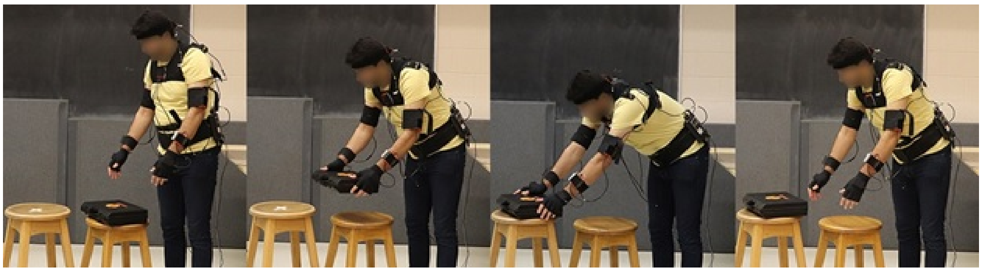


Figure 8.: A depiction of object manipulation task performed for kinematic data collection. The participants were expected to lift and lower an object between two target locations in the sagittal plane. The joint trajectories were measured using an upper body Xsens MVN suit (Xsens Technologies, Enschede, Netherlands).

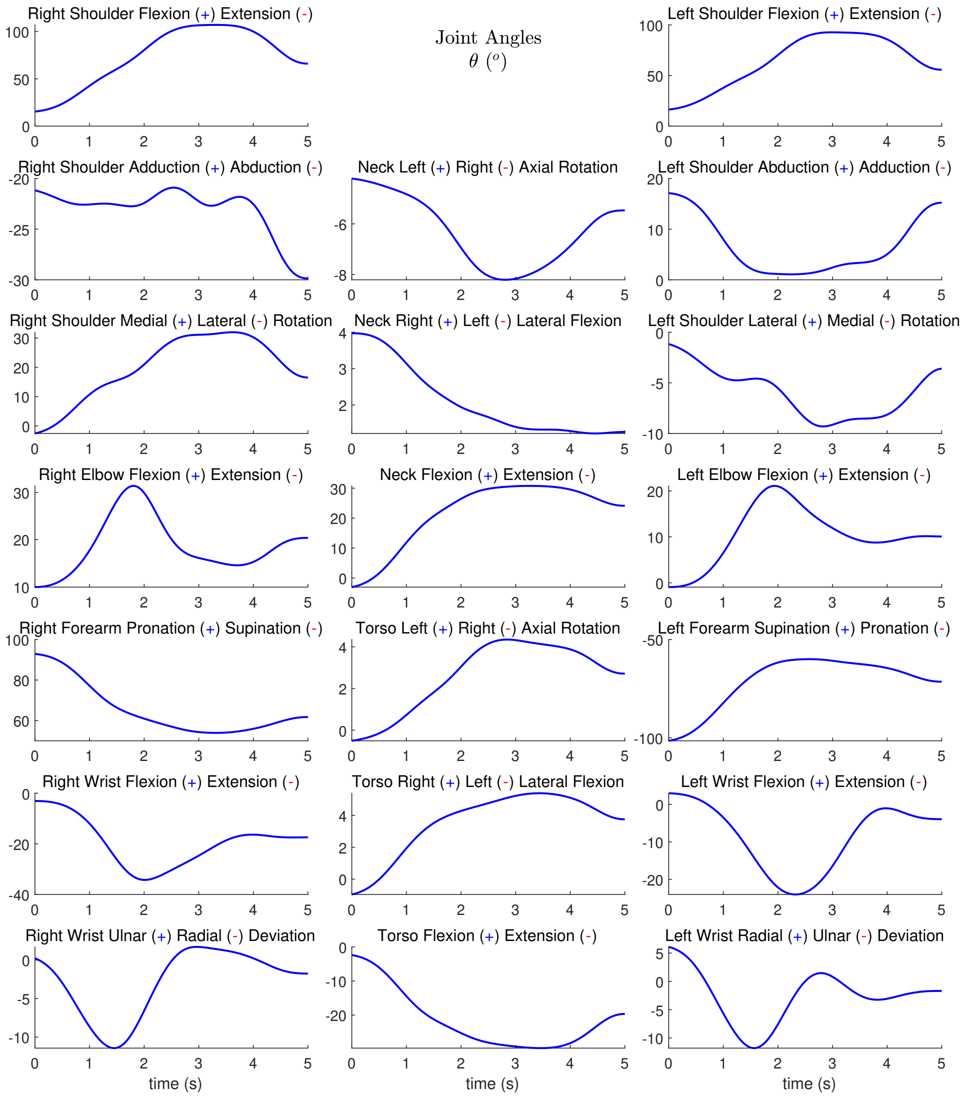


Figure 9.: The trajectories of 20 joint angles (^{o}) acquired from the motion recording process and 5 Hz low pass filtering.

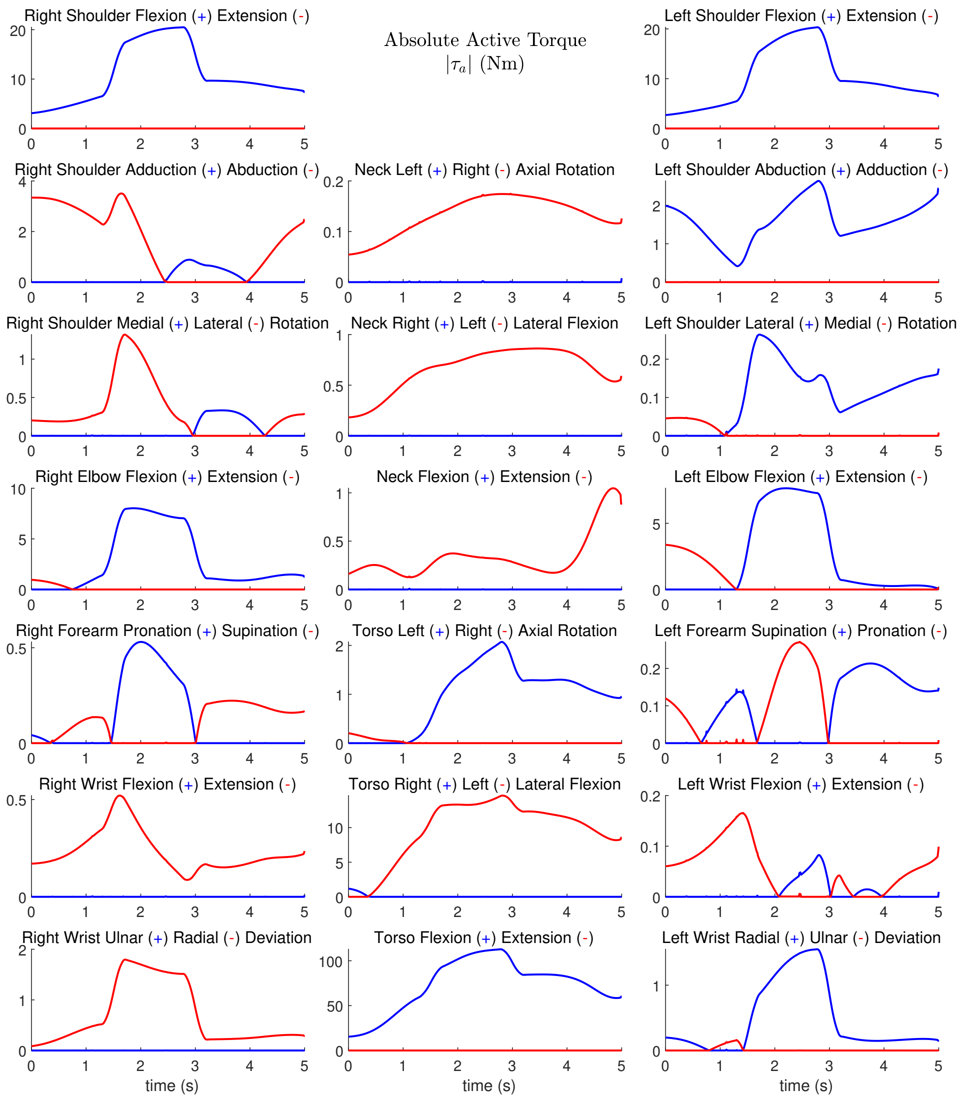


Figure 10.: The plot of absolute active torque of 20 joints and 40 MTGs optimized for minimum active torque in the pick-and-place movement.

muscle activation [106], metabolic energy [107], and muscle fatigue [108].

The CNS coordinates human movements with a complex signal that may be computed using (I) motion prediction and (II) corrective command [99]. An internal model or representation of the MSK system is used to compute motion prediction or feed-forward control. In addition, the sensory organs calculate a correction instruction or feedback control to remedy any mistakes caused by model uncertainty, external disruption, or an unfamiliar environment. Nonlinear model predictive control (NMPC) emulate this multi-purpose control with a limited horizon (feed-forward and feedback control) [99] (Figure 13). The NMPC employs an internal model (IM) that is learned over time [109, 110] to simulate human dynamics to forecast optimal motion (feed-forward) and correct prediction mistakes (feedback) [99]. It evaluates the ideal motion using a cost function. This cost function comprises joint angle errors $(\boldsymbol{\theta} - \boldsymbol{\theta}_\mathbf{d})$, joint angular velocity errors $(\boldsymbol{\omega} - \boldsymbol{\omega}_\mathbf{d})$, model inputs and their derivatives, as well as weights

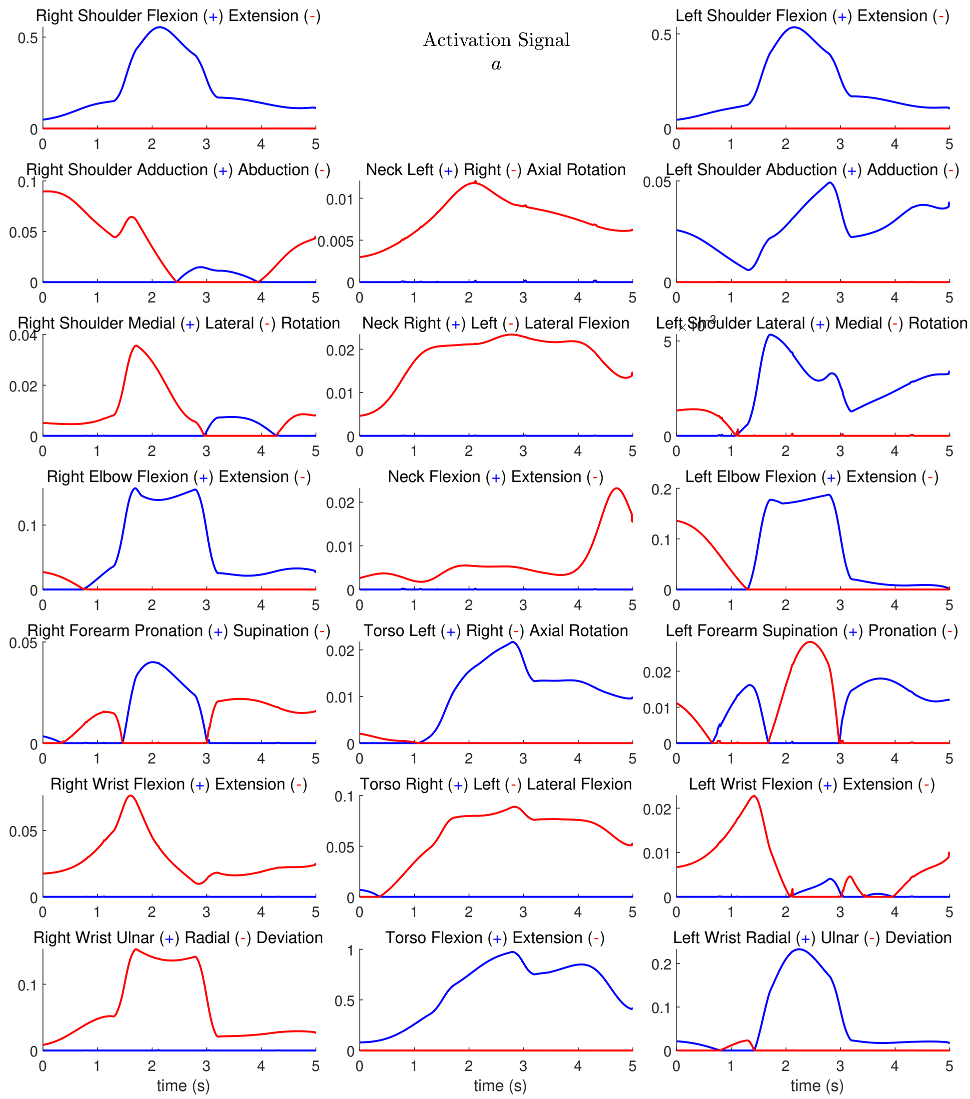


Figure 11.: The activation signal of the 20 joints and 40 MTGs for activation of the MTGs driving the optimized active torque in the pick-and-place movement.

(**W**). For the activation-signal-driven model (input $\boldsymbol{a}$) , this cost function can be formulated as Equation (15).

$$J = \int_0^{t_f} \left[\mathbf{W}_1^T \left(\boldsymbol{\theta} - \boldsymbol{\theta}_\mathbf{d}\right)^2 + \mathbf{W}_2^T \left(\boldsymbol{\omega} - \boldsymbol{\omega}_\mathbf{d}\right)^2 + \mathbf{W}_3^T \left(\boldsymbol{a}\right)^2 + \mathbf{W}_4^T \left(\dot{\boldsymbol{a}}\right)^2 \right] dt \tag{15}$$

$$\begin{gathered} \boldsymbol{\theta}_{min} \le \boldsymbol{\theta} \le \boldsymbol{\theta}_{max} \\ \boldsymbol{\omega}_{min} \le \boldsymbol{\omega} \le \boldsymbol{\omega}_{max} \\ \mathbf{0} \le \boldsymbol{a} \le \mathbf{1} \\ \dot{\boldsymbol{a}}_{min} \le \dot{\boldsymbol{a}} \le \dot{\boldsymbol{a}}_{max} \end{gathered} \tag{16}$$

Under the influence of system dynamics provided in section 5 and constraints in Equation (16), the NMPC controller optimizes the cost function (15).

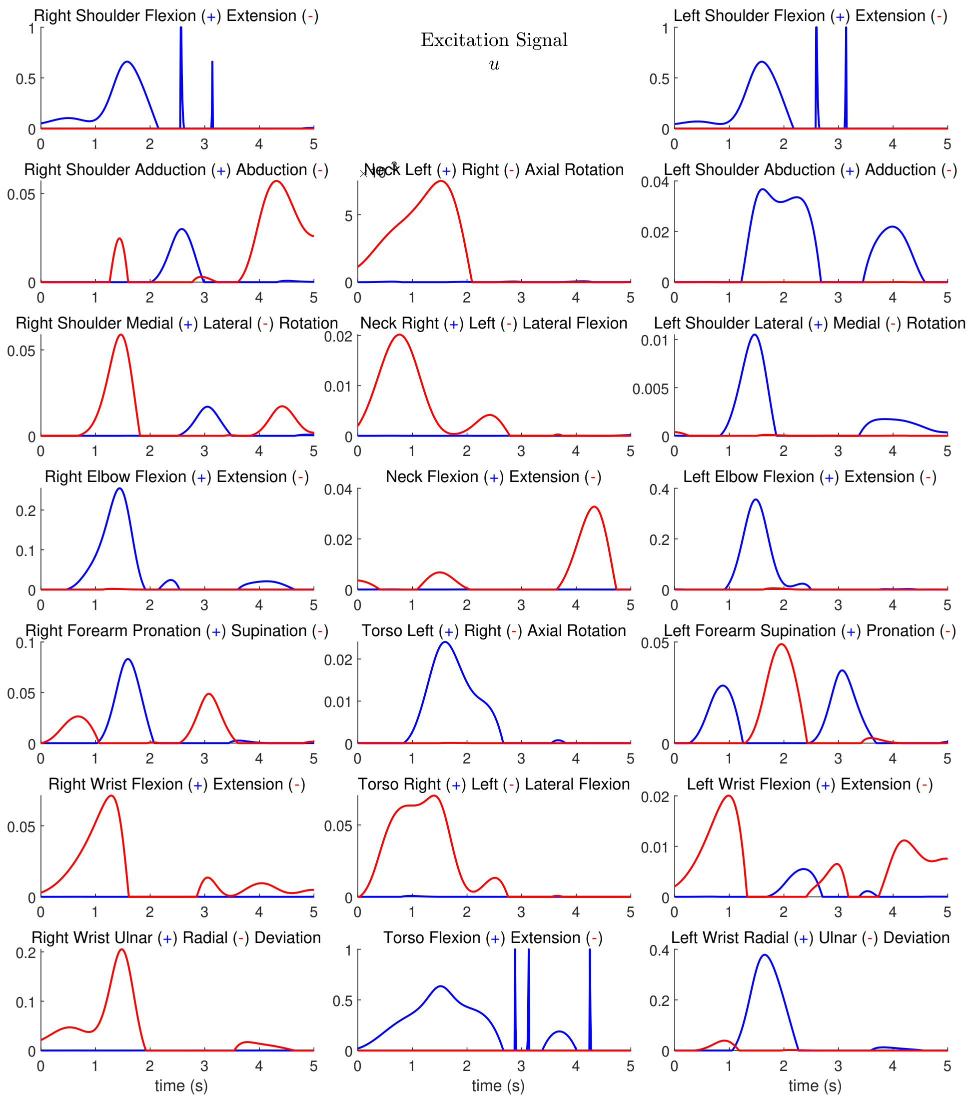


Figure 12.: The excitation signal of 40 joint MTGs for the pick-and-place movement.

The MTG excitations or activations in the musculoskeletal model may be generated by: (1) a forward static optimization (FSO), in which a musculoskeletal system behavior is approximated by static reduction of a physiological cost functional, or (2) by a forward dynamic optimization (FDO), in which a musculoskeletal system behavior is modeled through dynamic minimization of a physiological cost functional, or by (3) experimental sEMG data from a subject. The sEMG data may be mapped to MTG inputs with a machine learning model [87].

7. Model validation

We compare the measured moments at the joints of interest to the model's estimated ones to assess the model's performance for a variety of different predicted movements. The experimental study uses dynamotor measurement devices that constrain joint motion to measure joint moments. In these tests, subjects are directed to flex or stretch

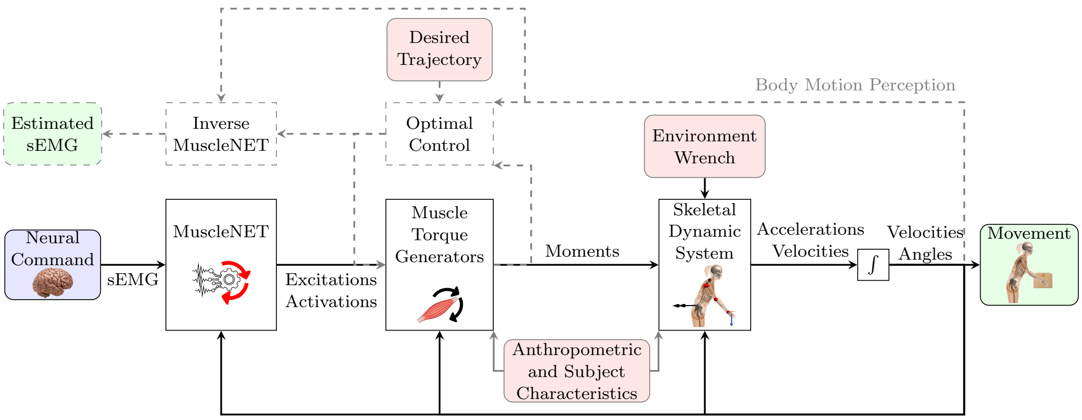


Figure 13.: Elements of a typical musculoskeletal simulation in MapleSim Biomechanics. Movement occurs from a complex composition of the neural, muscular, and skeletal systems. MapleSim Biomechanics comprises computational models of these systems, allowing for human movement prediction and analysis. Neural commands to MTGs, in the form of excitations, can be estimated from optimal controller models or experimental data (e.g., sEMG) with MuscleNET (a machine learning framework that maps EMG excitations to kinematic/dynamic variables) [87]. MapleSim Biomechanics's MTGs models, which convert excitations into biomechanical joint torque, contain the passive torque, active torque angle scaling, and active torque angular velocity scaling features of the biomechanical joint. MapleSim Biomechanics MTG are based on experimental data. MapleSim Biomechanics's peak joint strength depends on age, height, body mass, dominant side, and athletic activity.

maximally against the device's resistance, and the resulting isometric and isokinetic torques are recorded. The maximum isometric test is done with zero joint velocity. In contrast, the isokinetic test is done as the dynamotor measurement system controls the joint velocity. The isometric and the isokinetic tests are also simulated using the MapleSim Biomechanics model and forward dynamics. We consider maximum activation for MTG in the model when calculating maximum isometric moment-generating capability.

According to section 4.2, the joint peak isometric torque is a function of age [27, 29, 30], height [27, 31, 33], body mass [27, 31, 32], as well as, sex, dominant side [34, 36], and athletic activity [47].

7.1. Isometric test

The results of many published experiments are represented in the maximum isometric joint moments computed using the MapleSim Biomechanics model. Since the published experiments used for validation all had a limited number of subjects (less than 10), we focus on the trend and pattern of maximum isometric joint torques in our comparison with the experimental literature. Modeling and experimental results for the torso and neck are compared in Figure 14, for the shoulder in Figure 15, and for the elbow, forearm, and wrist in Figure 16.

The maximum isometric torques generated by torso flexors and extensors as a function of flexion angle successfully follow the trend of the data reported by Keller and Roy [54] and Kumar [24] in the ROM of torso flexion/extension (Figure 14a). Surprisingly, the data reported by Kumar [24] covers more than the average possible ROM that is measured by most researchers [52, 53, 55] for the torso joint. The maximum isometric moment produced by torso right and left flexors in Figure 14b also follows the pattern reported in the study of Kumar [24] for the ROM of torso lateral flexion [52, 56]. As shown in Figure 14.b, the torso left axial rotator isometric torque matches the data by Kumar [24] well, although Kumar [24]'s experiments did not cover the full ROM. The torso's right axial rotator's maximum torque decreases when the torso axial angle increases, which is similar to the experimental data of Kumar [24] (Figure 14c). Similar trends between the experimental data of Jordan et al. [31] and the estimated isometric torque of neck flexion and extension by the MapleSim Biomechanics model are shown in Figure 14d. The neck peak flexor torque is generated at approximately 60^o neck flexion angle by MapleSim Biomechanics, similar to the experimental data of Jordan et al. [31]. Other-DoF of neck has no experimental isometric torque versus neck joint angle data for validation.

The model's shoulder flexion moment-generating capacity diminishes at the same rate as in the literature [34, 62, 111] as shoulder flexion angle increases. The model's extension moment-generating strength follows the same pattern as the data in the literature [62], and our dynamometer experiment as mentioned in section 4.3, as shoulder flexion angle increases (Figure 15a). The model reproduces the empirical trend of literature data [34, 62] in shoulder vertical abduction and adduction strength as the shoulder abduction angle increases (Figure 15b). The vertical adduction torque peaks at approximately -60^o rotation similar to the experimental data of Garner and Pandy [62]. As the shoulder horizontal abduction angle changes, the model reproduces the experimental trend in shoulder horizontal abduction and adduction strength (Figure 15c). Horizontal abduction torque peaks at approximately -50^o rotation angle like the experimental data. The model's maximum isometric external/internal rotation moment follows the same pattern as that seen in the literature [34, 46, 61, 62] (Figure 15d).

As shown in Figure 16a, the model's elbow flexion and extension moment-generating capacity follows the same trend as the research in [62, 112, 113], as shoulder flexion angle changes. In addition, the maximum elbow flexion and extension isometric torque occur at about 70^o and 80^o, respectively, which are similar to the literature [62, 112]. The forearm pronation isometric torque estimation has the same trend as Hutchins's experimental data [64] and has a slight difference in fully supinated arm with other studies [62, 114] (Figure 16b). The forearm supination torque profile has a similar trend and value as most of the reported values [62, 64, 114]. The estimated wrist flexion is similar to Garner and Pandy [62] and Delp et al. [67] (Figure 16c). The estimated wrist extension by MapleSim Biomechanics is similar to other research, and the joint angle in which the estimated maximum torque occurs is close to the values reported in the literature [62, 64, 67]. The wrist ulnar and radial deviation torque is similar to some sources [64, 67] and has a different trend with only Garner and Pandy [62] (Figure 16d).

7.2. Isokinetic test

Maximum isokinetic joint moments computed using the MapleSim Biomechanics model is compared with experimental data reported in the literature (shown in Figure 17). The experimental data used in most references are from elite athletes [115, 116] that generate torque even for higher angular velocities than the normal subject re-

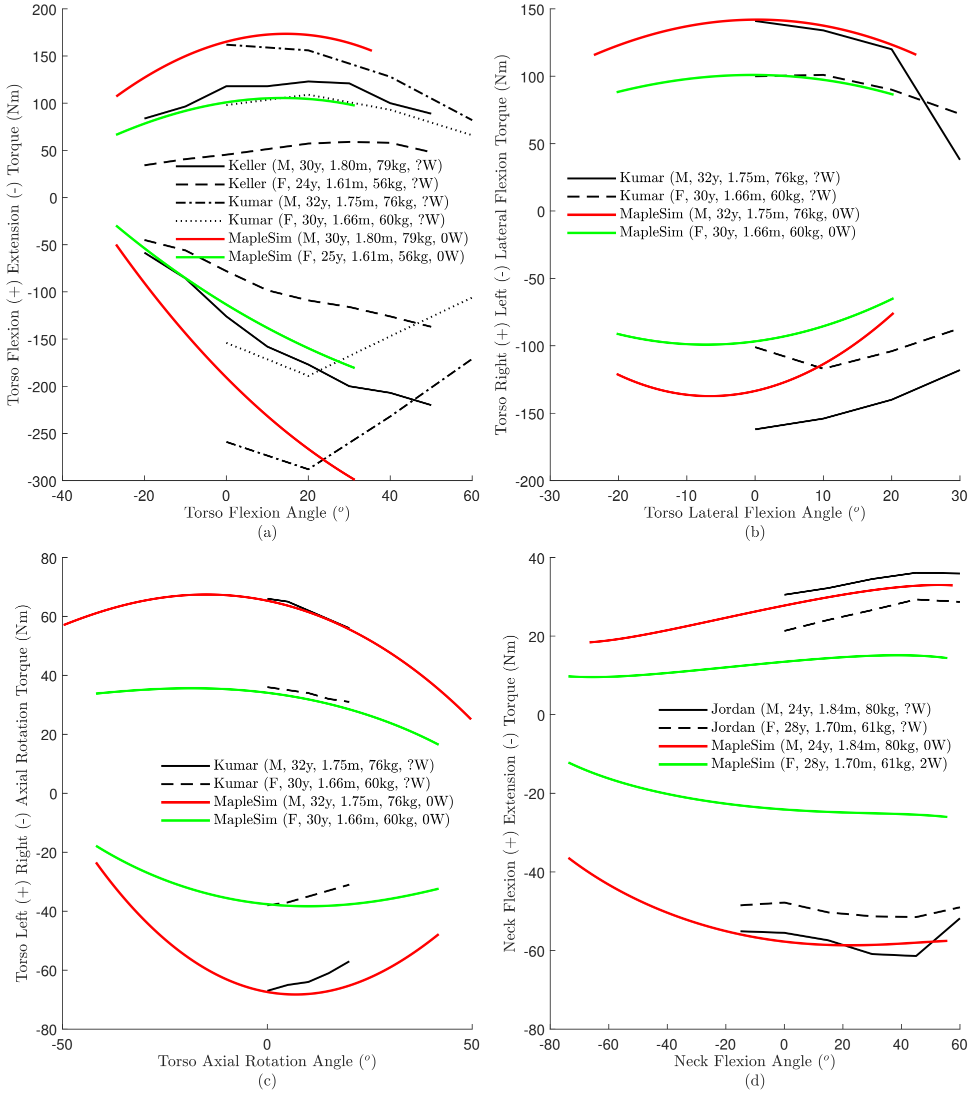


Figure 14.: Maximum isometric torque generated by (a) torso flexors (+) and extensors (-) versus torso flexion, (b) torso right (+) and left (-) lateral flexors versus torso lateral flexion, (c) torso left (+) and right (-) axial rotator versus torso axial rotation, and (d) neck flexors (+) and extensors (-) versus neck flexion. The model estimates are highlighted in thick solid lines and are compared to experimental data from Keller and Roy [54], Kumar [24], and Jordan et al. [31]. M: Male, F: Female, Height is in meter (m), Weight is in kilograms (kg), number of workout sessions per week showed with W.

ported in Table 4 are capable. In addition to not comparing the amplitude of the isokinetic torque, the maximum possible torque for the highest reported angular velocity is not compared in this research due to the difference in subject types.

As shown in Figure 17, the experimental torques decrease as the angular velocity increases for the concentric isokinetic test of joints, similar to the estimation of MapleSim Biomechanics [74, 115–118]. For the eccentric isokinetic test, the experimental torques increases as the angular velocity increases [116] similar to the MapleSim

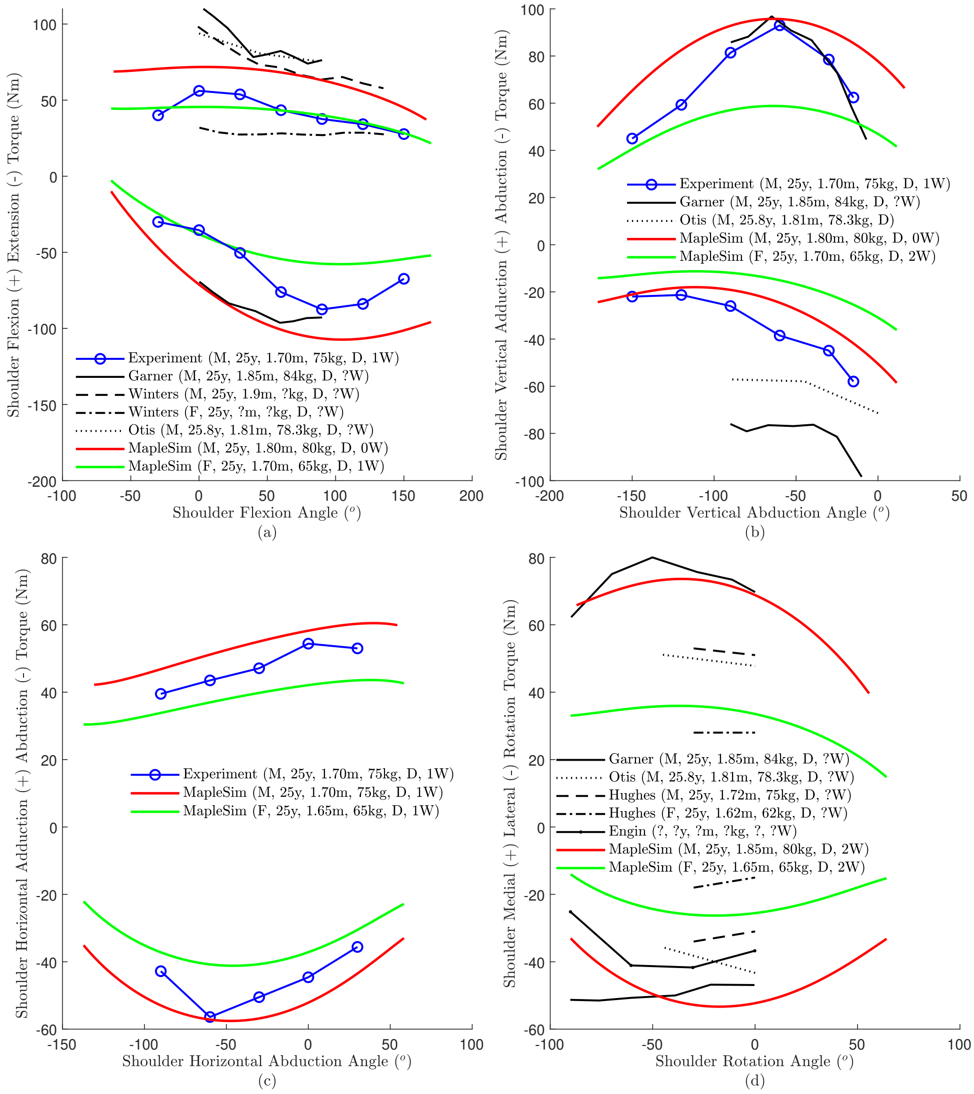


Figure 15.: Maximum isometric torque generated by (a) shoulder flexors (+) and extensors (-) versus shoulder flexion rotation, (b) shoulder adductors (+) and abductors (-) versus shoulder abduction rotation, (c) shoulder horizontal adductors (+) and abductors (-) versus shoulder horizontal abduction rotation, and (d) shoulder medial (+) and lateral (-) rotators versus shoulder rotation. The model estimates are highlighted in thick solid lines and are compared to experimental data from our test (mentioned in section 4.3), Garner and Pandy [62], Winters and Kleweno [111], Otis et al. [34], Hughes et al. [46], and Engin and Kaleps [61]. SFE, SVAA, and SHAA were determined with a straight arm in the neutral position. SMLR torque was determined for the elbow in 60^o flexion, and 60^o abduction. M: Male, F: Female, Height is in meter (m), Weight is in kilograms (kg), number of workout sessions per week showed with W.

Biomechanics simulation. In total, the maximum isokinetic torque of joints estimated by the MapleSim Biomechanics model follows the same pattern of the experimental data reported in the literature [74, 115–118].

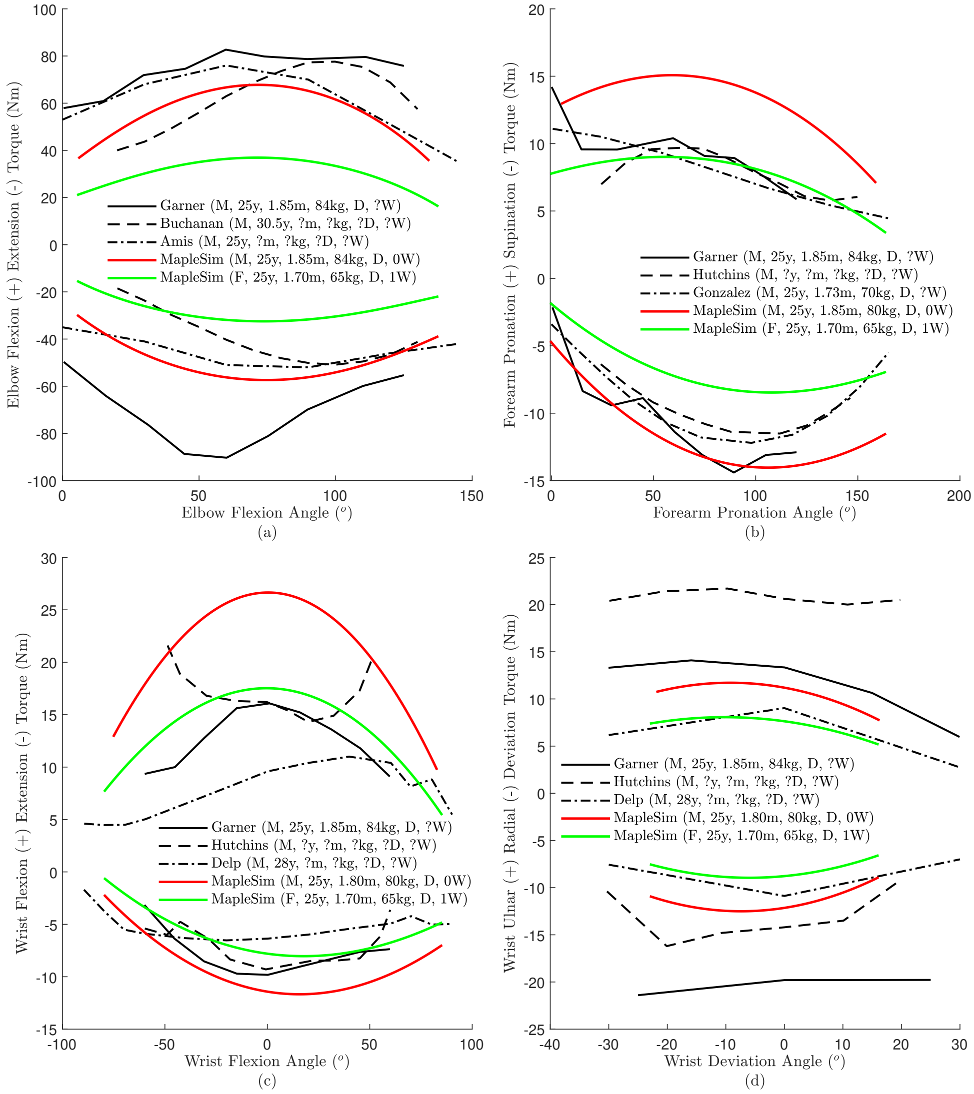


Figure 16.: Maximum isometric torque generated by (a) elbow flexors (+) and extensors (-) versus elbow flexion angle, (b) forearm pronators (+) and supinators (-) versus pronation angle, (c) wrist flexors (+) and extensors (-) versus wrist flexion angle, and (d) wrist ulnar (+) and radial (-) deviators versus deviation angle. The model estimates are highlighted in thick solid lines and are compared to experimental data from Garner and Pandy [62], Buchanan et al. [113], Amis et al. [112], Gonzalez [114], and Delp et al. [67]. EFE, FPS, WFE, and WRUD were determined with the shoulder in 60^o flexion, and 30^o abduction and all other DoF in neutral position. M: Male, F: Female, Height is in meter (m), Weight is in kilograms (kg), number of workout sessions per week showed with W.

7.3. Shoulder workspace volume

The shoulder workspace has not been evaluated in most biomechanical modeling studies. The ROM for the right shoulder joint of a male is shown in Figure 18. For calculating the ROM, the first and second shoulder Euler angles have been varied from $-\pi$ to π. Then, the passive torque of the shoulder joint has been calculated using the

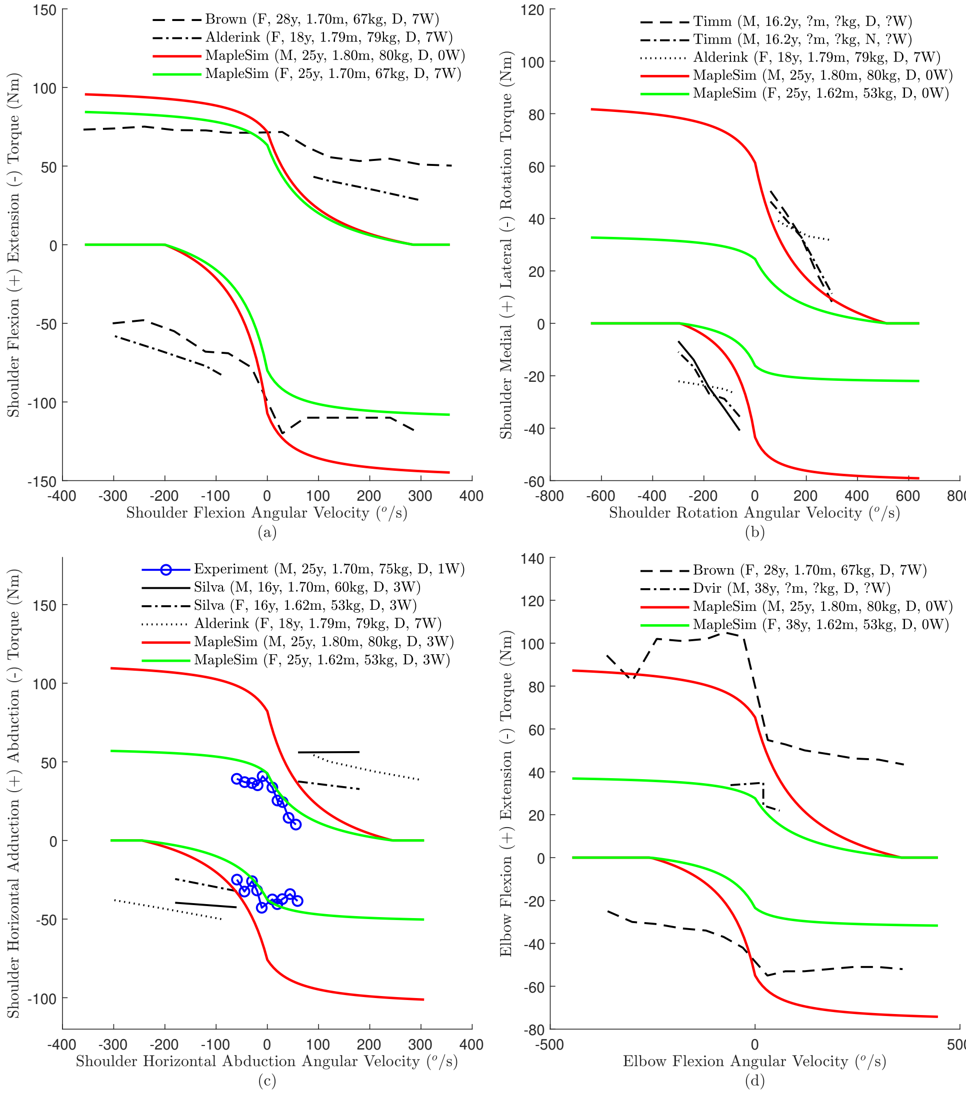


Figure 17.: Maximum isokinetic torque generated by (a) shoulder flexors (+) and extensors (-) versus shoulder flexion angular velocity, (b) shoulder medial (+) and lateral (-) rotators versus shoulder angular velocity, (c) shoulder horizontal adductors (+) and abductors (-) versus shoulder horizontal abduction angular velocity, (d) elbow flexors (+) and extensors (-) versus elbow flexion angular velocity. The model estimates are highlighted in thick solid lines and are compared to experimental data from our test (mentioned in section 4.3), Brown [116], Alderink and Kuch [115], Timm [118], Silva et al. [74], Dvir [117]. M: Male, F: Female, Height is in meter (m), Weight is in kilograms (kg), number of workout sessions per week showed with W.

2 MTG models of SFE and SAA. As described in section 4.6, for the second-DoF of the shoulder (SAA), the extrapolation of 2 MTG models of the SVAA and SHAA has been used. The ROM that has less passive torque than 5 Nm has been visualized in Figure 18. As described in section 4.1, the passive torque of the joint outside of the ROM increases exponentially. Thus, the 3D workspace in Figure 18d shows the reachable surface of the upperarm, which is quite similar to the 3D experimental data

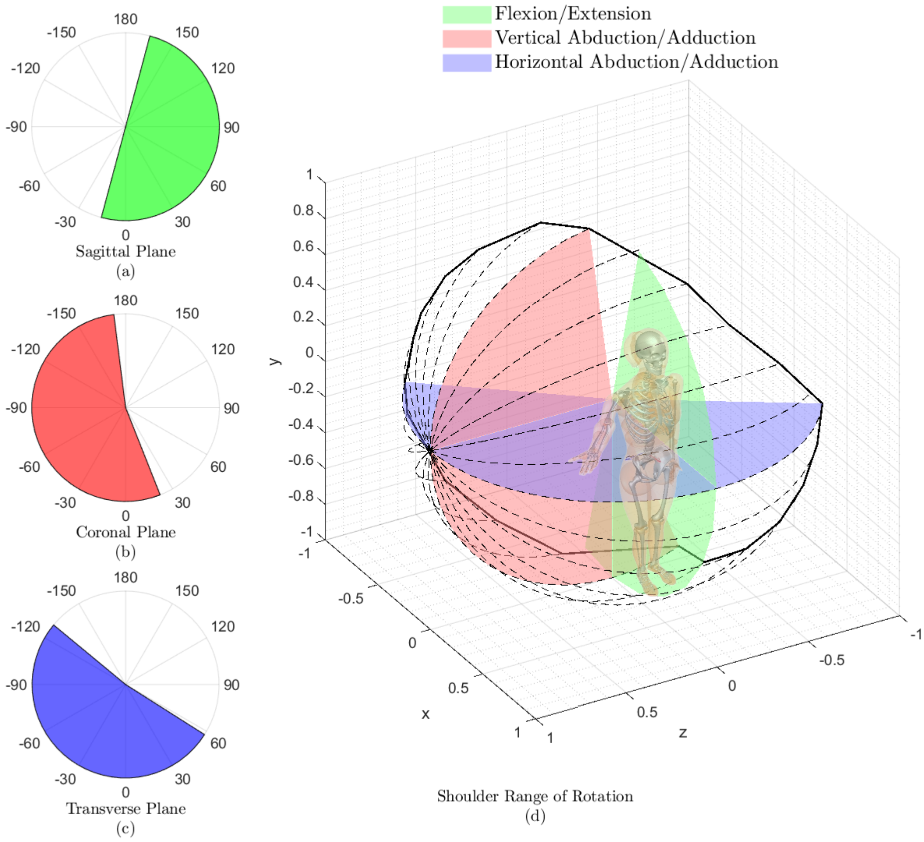


Figure 18.: The visualization of a male right shoulder ROM for (a) Flexion/Extension from the side view, (b) vertical abduction/adduction from the front view, (c) horizontal abduction/adduction from the top view, and (d) 3D reachable surface of a unit (1-meter length) upper-arm.

in [119, 120] and 2D experimental data in the literature [52, 60].

8. Discussion

This paper presents a musculoskeletal model for torso, neck, shoulder, elbow, and wrist joints in the human upper body. The model is based on average experimental biomechanical data for segment properties (dimensions and inertia) as well as joint properties (maximum joint torque, maximum ROM, maximum angular velocity, and maximum isometric torque-angle profile). It can be adjusted based on subject body characteristics: sex, age, body mass, height, dominant side, and physical activity.

One of the contributions of this work is that it provides enough information to build a mathematical model. The data in Table 3, Table 4 and Table 8, as well as the presented equations, are sufficient to construct a dynamic model. Future users can focus on the targeted joint and ignore the non-targeted joint data.

To date, our model is more general and more complete in terms of being scalable with more anthropocentric variables than other presented MSK models; it is also directly verified and validated. The data for this general model is acquired from literature with multiple subjects in contrast to the subject-specific models in [20, 21, 62]. In contrast with most musculoskeletal models that are not validated for all ROMs [24, 34, 46, 61, 62, 111], our model was constructed and verified within the maximum ROM from the literature [52, 57, 60, 63, 66]. The average maximum angular velocity from the literature [53, 55, 56, 58, 59, 65] is used to limit the maximum isokinetic joint torque in concentric motion and verified against the experimental literature [74, 115–118]. The maximum joint torque is adopted from the literature for females and males [24, 27, 35, 46, 73–77]. The isometric joint torque-angle pattern is modeled from the literature data [24, 31, 54, 61, 62, 64, 67] or our own experimental measurement. The isokinetic torque-angular-speed profile is modeled using the literature data of [10, 44, 81, 82, 85] in addition to the mentioned average maximum angular velocity. Since the shoulder abduction/adduction has different parameters in different orientations, we have created a novel 2D MTG model, in contrast to previous musculoskeletal models of the upper limb [34, 62, 111], and verified it in two vertical and horizontal planes. Verification of the isokinetic test of each joint, which is essential for simulation of high-speed motion (mostly sports analysis), is missing in previous musculoskeletal models of the upper limb [62, 67, 112–114]. The workspaces of the shoulder flexion/extension and abduction/adduction are evaluated and verified with the literature [52, 60, 119, 120], which is essential for a complex motion of the shoulder.

The skeletal dynamic model can be adjusted for sex, body mass, and body height since they impact the anatomical locations, the COMs, and inertia parameters [25, 26]. The biomechanical joint torque is also a function of subject characteristics. The changes of peak torque relevant to age [27, 29, 30], height [27, 31, 33], body mass [27, 31, 32], dominant side [34, 36], and amount of activity [47] are modeled using the literature data. The maximum joint ROM can change according to sex differences in passive torque function. In addition, age impacts the excitation-to-activation delay, according to [91].

8.1. Applications

The provided biomechanical model can be used in various applications broadly categorized as: simulation of human movements (joint torques, ROM), biomechatronics, ergonomics and usability studies, predictive simulation of activities of daily living (reaching, walking, sitting, etc), clinical practice, human-device interactions, sports engineering, rapid and safe development/control of medical and assistive devices. Some of the mentioned applications are discussed in detail below.

Subject-specific torque-driven computer models of human movement have provided insight into optimal athletic technique [121]. Computer simulation is a valuable tool for investigating the athletic dynamics without repeating actual experiments [41]. These predictive simulation trials can provide insight into issues like equipment design [10], injury prevention [79], and optimum motor control methods [99], all of which increase sports performance [41], by providing researchers with an interface through which significant circumstances and parameters can be modified. Multibody computer simulation has been used for various activities (golf, track cycling, etc.) to study athletic performance [41].

Predictive multibody dynamic simulations are typically useful for rapid and safe development of new medical and assistive devices. Recent applications have included orthoses, prostheses, and wearable exoskeleton robotics [2, 17]. Models that include human interaction with biomechatronic devices (exoskeletons, prostheses, rehabilitation systems, etc.) are required for their design and control [5, 6, 17, 122]; these models are utilized within computer simulations [4, 123]. In addition, the human biomechanical model can be used for model-based control of robotic devices [4, 13, 87].

8.2. Limitations

Not providing information about muscle tension is the main weakness of the MTG models [2]. Anatomically detailed muscle models are needed to evaluate joint reaction force/torque as well as the muscle tensions. Our model uses MTGs instead of an anatomical-detailed model because of their ability to mimic the muscle group torque impact on the targeted joint. Note that without direct verification of parameters for the anatomical-detailed model, the estimated muscle tension cannot be trusted. Our model has a wide range of applications in sports and biomechatronics analysis, but is limited for from muscle injury analysis.

Some parameters are not provided in the literature for all the joints; thus we considered the same constant value for all the them. In the future, measuring this information with a high number of subjects can increase the accuracy of our model, for example, the ratio of the maximum eccentric isokinetic torque (τ_m) over the maximum isometric torque (τ_0) and the impact of workout on the maximum joint angular velocity.

Using this model in EMG-driven simulation suffers from a gap of EMG to MTG excitation mapping, as shown in Figure 13. Having experimental EMG data for most upper body muscles can create new opportunities for developing a machine learning mapping according to the MuscleNET model [87, 124].

The joint torque is modeled with a monoarticular assumption [68] of muscles, which is not always accurate. Experimental investigation of the biarticular muscles and the impact on the MTGs model can increase the accuracy of these models [45, 93, 121].

9. Conclusion

In this research, we created an adjustable MSK upper body model. The model has 20 DoF driven by 40 MTGs and is constructed from experimental literature data. The model is adjustable for different anthropometric measurements and subject body characteristics: sex, age, body mass, height, dominant side, and physical activity. The maximal isometric and isokinetic torque and workspace were compared with previously published research for model validation. This model contributes to sports analysis simulation, ergonomics and usability studies, predictive simulation of activities of daily living, human-device interaction evaluation, design, development, and control of medical, assistive devices, exoskeletons, and prosthesis.

Abbreviation(s)

BSIP body segment inertial parameters.
CNS central nervous system.
COM center of mass.
DoF degree of freedom.
EEG electroencephalography.
EFE elbow flexion/extension.
EMG electromyography.
FDO forward dynamic optimization.

FPS forearm pronation/supination.
FSO forward static optimization.
IM internal model.
IMU inertial measurement unit.
ISB International Society of Biomechanics.
MSK musculoskeletal.
MTG muscle torque generator.
NFE neck flexion/extension.
NMPC nonlinear model predictive control.
NRLAR neck right/left axial rotation.
NRLLF neck right/left lateral flexion.
ODE ordinary differential equation.
ROM range of motion.
SAA shoulder adduction/abduction.
sEMG surface electromyography.
SFE shoulder flexion/extension.
SHAA shoulder horizontal adduction/abduction.
SMLR shoulder medial/lateral rotation.
SO static optimization.
SVAA shoulder vertical adduction/abduction.
TFE torso flexion/extension.
TRLAR torso right/left axial rotation.
TRLLF torso right/left lateral flexion.
WFE wrist flexion/extension.
WRUD wrist radial/ulnar deviation.

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Notes on contributor(s)

A.N. and J.M. conceived and designed the content structure. A.N. and J.M. read and selected the references. A.N. and A.H did the data gathering using a dynamometer system. A.N. coded the models, simulations, and did validations. A.N. wrote the first draft, A.H. polished the draft. A.N. and J.M. revised the manuscript. J.M. secured funding for this research. All authors reviewed the manuscript.

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